

Risk Management Externalities in Agrifood Supply Chains *

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Abstract

Firms may under- or over-invest in risk management from the social planner's perspective, resulting in a negative externality on other economic agents in the supply chain. The externality of risk management is particularly relevant for agrifood supply chains, which frequently experience shocks and play a primary role in preventing major social losses from food insecurity. The magnitude of the externality depends on the structure of the supply chain. We build a theoretical model to characterize such externalities in US food supply chains where intermediary firms choose output quantities and investments in managing risks. The model allows for flexible market structures and interdependence of risk management among firms. Risk interdependence captures the unique feature of biotic hazards (e.g., animal and plant diseases) in agrifood supply chains, where the effectiveness of a firm's risk management depends on peer firms' behavior. We offer novel insights on the role of risk interdependence in driving the externality in risk management under different market structures. We show that private firms invest less than the socially optimal level under perfect competition, but risk interdependence and market power introduce complex incentives in risk-reducing investment that shape the externality. In particular, when investments are complements across firms, and firms possess strong market power in production but limited coordination in risk management, private investment may equal or even exceed the socially optimal level.

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Supply chain disruptions have become increasingly frequent in the United States and globally (Baldwin and Freeman 2022). In response, governments have discussed and implemented a variety of policy interventions, attempting to make supply chains more resilient to shocks (Elliott and Golub 2022; Grossman, Helpman, and Sabal 2024). Agrifood markets, in particular, have garnered the attention of policymakers because recent disruptions like extreme weather and disease outbreaks have resulted in stock-outs and elevated prices of staple food products (Hobbs 2021; Hobbs and Hadachek 2024). These events affect all agents along the supply chain, including agribusiness firms that can endogenously reduce their exposure to hazards through risk management. A critical open question for policymaking is whether private firms arrive at the socially optimal level of risk management on their own. If not, how large is the externality, and what policy intervention is warranted?

To characterize the externality in risk management in the context of the US agrifood sector, we build a theoretical model of a vertical supply chain where intermediary processing/retailing firms are exposed to hazards that threaten their ability to produce or deliver the final products to consumers. Firms may invest in risk management to reduce exposure to hazards. The model allows for 1) a flexible market structure and 2) interdependence in risk management among firms (e.g., the effectiveness of vaccines for animal diseases on a farm supplier depends on actions by peers), both of which are key features of the US agricultural sector (Kunreuther and Heal 2003; Hennessy, Roosen, and Jensen 2005; Sexton 2013). Because data on firms' costs and decisions are proprietary and unavailable to researchers, we calibrate the model using the existing empirical literature on biotic hazards (biohazards hereafter) in US agrifood markets. We then quantify the externality under various market and risk conditions and measure the welfare implications of policy intervention.

In the model, profit-maximizing firms in the middle of the supply chain anticipate hazards and can endogenously reduce their exposure to the hazards by investing in risk management before the realization of shocks.¹ At the same time, each firm chooses the output level to maximize its own expected profits. With the two choice variables, firms may collude (explicitly or tacitly)

in production — the conventional force behind market power — and in risk management. The model’s two key innovations are, first, allowing collusion in risk management itself, and second, allowing the effectiveness of one firm’s investment to depend on peer firms’ investments.

To compare firms’ choices against the social optimum, we consider an alternative scenario in which the social planner manages risk. The social planner first chooses the level of risk-reducing investment by firms that maximizes total social welfare (i.e., consumer surplus, producer surplus, and firm profits). In the second stage, firms choose profit-maximizing outputs given the risk environment set by the planner. This setup captures the realistic context where the government imposes risk management standards (e.g., requiring vaccines to prevent animal diseases or biohazard protocols), but allows the market to dictate equilibrium outputs and prices.

In the absence of data to empirically test the model’s predictions, we draw parameter values from distributions based on empirical literature on biohazards and US agrifood markets. We perform Monte-Carlo simulations under counterfactual risk and market structure scenarios to demonstrate the externality in risk management and welfare gains from policy intervention. The baseline scenario considers a simple context without collusion among firms or interdependence in risk management. Simulations reveal a positive *public-private wedge in risk management*: Firms invest less than the social optimum, and the wedge plateaus once the hazard is sufficiently large. Intuitively, this wedge reflects a risk management externality that arises because private firms fail to incorporate the welfare impacts caused by the propagation of disruptions in food supply on the greater society (i.e., consumers and farmers) and, thus, under-invest in risk reduction from the social planner’s perspective ([Baldwin and Freeman 2022](#)).

Animal and plant diseases also fall into a class of risks with an *interdependent security* problem ([Kunreuther and Heal 2003](#); [Acemoglu et al. 2012](#)), where one firm’s incentive to protect against risk depends on other firms’ behaviors. Risk investments can be substitutes or complements among firms ([Wang and Hennessy 2015](#)). For example, if the substitution among firm investments is not accounted for due to low coordination in risk management, the firm free-rides and tends to invest less than in a setting of independent investment, widening the risk externality. Simulations

suggest that potential welfare gains vary considerably with the degree of investment interdependence, highlighting the contextual importance of risk interdependence for policymaking.

Next, we study the interactions between market power and the externality by imposing flexible assumptions on competition in the style of the Flexible Oligopoly/Oligopsony Market (FOOM) model (e.g., [Russo, Goodhue, and Sexton 2011](#)). Market power is characterized by the conjectural variations in input procurement (i.e., the conventional buyer power), final output (i.e., the conventional seller power), and a risk conjectural variation. Intuitively, the conjectural variation reflects the degree of firm collusion in conducting an action: a higher degree of collusion implies greater market power exhibited by firms. We show that the three forms of market power play critical and sometimes opposing roles in determining the sign and magnitude of the risk management externality. These findings offer insight into how federal agencies, such as the Department of Justice and the Federal Trade Commission, might factor in supply-chain resilience trade-offs in antitrust policy cases.

Our model highlights multiple economic forces that shape output and risk management decisions jointly. These forces differ between private firms and the social planner, giving rise to a public-private risk investment wedge, and their relative strength varies with market power. First, firms only capture a share of social welfare and have weaker incentives to mitigate hazards than the social planner, who also values the gains from risk reduction accruing to consumers and producers. As market power increases, firms obtain a greater share of social welfare. This divergence between private and social incentives narrows with market power, and the investment wedge shrinks toward zero. We term this the *social welfare effect*.

Second, investment affects industry-wide effective supply and profit margins. When firms fail to coordinate in investment choices, each firm does not fully internalize the negative effect of its own investment on peers' profits through lower equilibrium margins. When market power makes margins more valuable, this can generate excessive private investment relative to the planner, who effectively coordinates firms' risk management by setting the firm-level investment. We refer to this force as the *investment coordination effect*.

Third, investment affects output indirectly because it enters marginal production costs. Higher investment can reduce equilibrium output by raising per-unit costs and depressing output. Investment can hence serve as an indirect restriction on effective supply, incentivizing firms to invest more for monopoly/monopsony profits and creating an *indirect output effect*. When buyer and seller power are sufficiently strong, the coordination and indirect output effects may dominate the welfare effect, causing firms to over-invest in risk management relative to the social planner.

We mainly contribute to the discussion on resilience to shocks and risk management in supply chains (Barrot and Sauvagnat 2016; Carvalho et al. 2020). Our study is one of the first to offer a flexible conceptual framework that characterizes firms' risk management decisions, unpacks the driving mechanisms, and weighs the equilibrium values against socially efficient risk management in the context of US food supply chains. We provide theoretical evidence on the externality of managing risks by firms and the corresponding impacts on welfare along the supply chain under various market and risk conditions.

We also contribute to the literature that explores the implications of market structure for agrifood supply chain resilience. Past work has empirically measured how market structure (Gnutzmann, Kowalewski, and Śpiewanowski 2020) and diversification in production/sourcing (Stevens and Teal 2024) affect economic outcomes in the presence of historical shocks. Other theoretical work has measured how market power (Hadachek, Ma, and Sexton 2024) or flexible marketing options (Manfredo et al. 2025) can make agrifood supply chains more resilient. Our findings speak directly to the management of biohazards in agrifood supply chains and the role of agencies like the USDA Animal and Plant Health Inspection Service (APHIS) in preventing social welfare losses.

In particular, risk management taken by an agent in the supply chain may inadvertently affect the risk exposure of other agents, due to the nature of how biohazards are transmitted and spread (Wang and Hennessy 2014). We show that the nature of this risk dependence has critical implications for the size of the risk management externality. In practice, the interdependent nature of biohazard management may also increase the likelihood of coordination between firms (Kunreuther and Heal 2003; Flowers, Kaplan, and Singh 2025) and government intervention (Hennessy

and Wolf 2018).² Our findings are among the first to highlight the interactions of these factors on risk management decisions and to quantify the welfare consequences of firm-driven decisions relative to the social optimum.

More broadly, our discussion of policy intervention in interdependent risk management speaks to the regulation of systemic risks in many non-agricultural supply chains. For example, governments and regulatory agencies intervene in electricity markets through regional transmission organizations and independent system operators to coordinate reliability and manage externalities arising from interdependent electricity grids (Joskow 2024). Similar concerns arise in banking and cybersecurity regulations (Asghari, van Eeten, and Bauer 2016), where decentralized private incentives may not align with system-wide resilience objectives due to externalities and interdependence. Our key insight that the optimal design of risk-management policies depends on market structure and the nature of interdependence across firms applies in these settings.

Biohazards in Agrifood Supply Chains

Agribusiness firms manage a long list of abiotic and biotic hazards that affect their business (Oerke 2006), and they invest in programs or operating inputs to mitigate exposure to such hazards. Regarding many abiotic hazards (e.g., natural disasters, mechanical plant fires, geopolitical disruption), the returns to risk management accrue solely to the decision-making firm, and an individual firm's decisions do not have obvious spillovers on other firms' risk exposure. Managing biotic factors introduces more complex relationships. In particular, biohazards are known to be interdependent — risk investments between firms can be substitutes or complements to each other due to the transmission of disease or the mobility of pests (Hendrichs et al. 2007; Hutchison et al. 2010; Wang and Hennessy 2015).

Every year, the US agricultural sector suffers substantial losses due to plant and animal diseases caused by pests and pathogens. For instance, crop losses due to plant pathogens cost farmers an estimated \$21 billion annually (USDA 2021). In response, the US agricultural sector,

both public and private entities, invests substantially each year in preventing and managing plant and animal diseases. Every year, farmers spend more than \$20 billion dollars (about 5% of farm production expenditure) on agricultural pesticides, fungicides, and animal vaccines ([USDA 2023](#)).

Plant diseases Pests are organisms, including insects, mites, nematodes, and pathogens that can damage economically valuable agricultural crops. In U.S. crop production, they cause hundreds of millions of dollars in annual losses through reduced yields and increased chemical and labor costs ([Fan et al. 2020](#); [Fernandes-Cornejo and Jans 1999](#)).

Citrus greening is a recent major plant disease caused by the Asian citrus psyllid. The disease was first detected in Florida in 2005 and spread to California, Texas, and other citrus-producing states over the last 20 years. Infected trees exhibit yellow shoots, blotchy-mottled leaves, lopsided and sour fruit, premature fruit drop, and canopy thinning. Trees typically decline and die within two to five years of infection. From 2006 to 2016, the disease resulted in an estimated loss of \$9 billion in Florida alone ([Fuchs et al. 2021](#)). The interdependent nature of citrus greening spread is the primary reason why public or collective action efforts are in place to impose uniform standards for growers ([Singerman, Lence, and Useche 2017](#); [Singerman and Rogers 2020](#); [Lence and Singerman 2023](#)). For example, [Flowers, Kaplan, and Singh \(2025\)](#) show that citrus growers in California have engaged in collective risk management strategies, like area-wide pesticide application.

Another example is Fusarium head blight in wheat, also known as scab. Scab causes lower yields and grain quality and is a perennial management challenge for wheat farmers. The disease is primarily caused by a fungus that infects wheat heads during flowering, especially under warm and humid conditions. Infected wheat kernels become shriveled and discolored, often exhibiting an orange-pink hue. In 2019 alone, the northern United States experienced over 45 million bushels of wheat yield losses due to the disease ([The Crop Protection Network 2022](#)). Crop rotation and fungicide applications are key management responses taken by producers. Like farmers protecting against citrus greening, wheat growers promote region-wide communication, joint field trials, and

shared data systems to help prevent the fungus from harming their crops ([McMullen et al. 2012](#)).

Animal diseases Infectious animal diseases carried by bacteria and viruses impose constant threats to input sourcing by meat processors. For example, blue ear disease in swine, bird flu, swine flu, and foot-and-mouth disease in cattle have all been prominent in recent decades. Many outbreaks result in considerable losses to animal herds, causing supply shortages and long-lasting impacts in the corresponding industries ([Ma, Delgado, and Wang 2024](#)).

For instance, the United States experienced a highly pathogenic avian influenza (HPAI) outbreak from December 2014 to June 2015. Producers euthanized approximately 48 million chickens, turkeys, and other poultry as a result of the disease outbreak. [Greene \(2015\)](#) estimated the total economic losses, including costs of herd depopulation, restocking, and lost future production, at \$3.3 billion. Iowa and Minnesota bore 87% of the total damages ([Ramos, MacLachlan, and Melton 2017](#)). In early 2024, HPAI spread to dairy cattle as well, and subsequent cases of dairy herd infection were identified in at least 17 US states.

A Porcine Epidemic Diarrhea Virus (PEDv) outbreak occurred in the US hog industry in 2013. Within a year of the emergence of the disease, it had spread to 31 states and led to a 5-7% (about 7 million head) loss in US hog production ([Crawford et al. 2015](#)). PEDv is estimated to cause a \$0.9-1.7 billion annual loss to US swine producers alone ([Langel et al. 2016](#)). To help prevent the spread of the disease, the USDA made PEDv a reportable disease in 2014 and mandated tracking of pigs, vehicles, and equipment leaving the impacted places. The USDA further announced a \$5 million transfer to the industry to help with disease surveillance measures, movement tracking, herd monitoring, and epidemiological and research support for the development of a vaccine ([American Association of Swine Veterinarians 2014](#)). Due to the highly contagious nature of PEDv, biosecurity programs and data sharing networks have been formed among producers, sometimes with support from the government.³

Risk Mitigation by Agribusiness Against Biohazards

Although the protective actions against biohazards tend to be performed on farms (e.g., immunization of livestock), processing firms also play critical roles. Agribusinesses that buy raw farm outputs often enforce production and risk management protocols via formal and informal contracts with producers, especially in the highly contractualized livestock sector ([Hennessy 2005](#); [Crespi and Saitone 2018](#)). Processors effectively factor losses due to biohazards and the costs of protective actions on farms into the prices paid to farmers. Therefore, in supply chains with vertical production contracts, processors effectively bear the potential losses and the costs of risk management ([Spalding et al. 2023](#)).⁴

Given the realities of modern agrifood supply chains, we focus on a theoretical framework where the intermediary firms' optimizing actions dictate equilibrium outcomes at every stage of the supply chain, including the level where biohazards often materialize and are managed. This is consistent with the well-established literature of agrifood supply chains that models decisions from the intermediary firm's perspective (e.g., [Gardner 1975](#); [Schroeter 1988](#); [Russo, Goodhue, and Sexton 2011](#)).

Model

To compare the investment in risk management by intermediary firms and that of a social planner, we first study a private equilibrium in which a firm jointly chooses output and investment to maximize expected profits under a risk of shutdown. The probability of shutdown is endogenous and can be reduced through investment in risk management. The framework allows the firm to exercise both buyer and seller power. We model the social planner's problem as a two-stage game. In the first stage, the planner sets firms' investment levels to maximize expected social welfare; in the second stage, each firm chooses outputs to maximize its expected profits given the social planner's level of investment. We show that the resulting public–private wedge in investment depends on the degree of coordination among firms in output as well as investment and the cross-firm

interdependence in risk management.

Consider a two-stage supply chain with M farms and $N < M$ homogeneous processors/retailers that provide perishable food to consumers.⁵ Consistent with past supply-chain models (Gardner 1975; Sexton 2000), we assume an integrated processing-retailing sector for simplicity. We impose fixed proportions in production throughout the supply chain, such that a constant volume of the farm product is required to produce a unit of the consumer good. Therefore, we denote the equalized output produced at each stage of the supply chain by Q given the appropriate scale of units.

We specify the upward-sloping, inverse supply function of farmers as:

$$(1) \quad P^f(Q) = S(Q|X),$$

where P^f captures farm price, Q is market quantity, and X denotes exogenous supply shifters. Similarly, the downward-sloping, inverse consumer demand for the processed product is:

$$(2) \quad P^r(Q) = D(Q|Y),$$

where P^r is retail price, Q is market quantity, and Y captures exogenous demand shifters.

Each risk-neutral firm faces a common hazard – such as animal disease – that may restrict its ability to process and sell products. The hazard itself is exogenous, but its realized probability and severity are partially endogenous: firms can reduce their exposure by investing in risk-reducing technology (e.g., vaccination).

Denote the investment per output in risk management by I_j for firm j . Then, $\phi_j(I_j; I_{\neq j})$ equals the firm's actual probability of experiencing a shock. We allow individual investments to be technical complements or substitutes among firms to reflect a key attribute of biohazards. Specifically, when firm investments are complements (substitutes), $\frac{\partial^2 \phi_j}{\partial I_j \partial \bar{I}} < 0$ (> 0) with $\bar{I}$ being the average investment of peer firms in the industry. Intuitively, this means that the marginal risk reduction from increasing I_j is larger if a firm's own investment, I_j , and other firms' investments,

$\bar{I}$, reinforce each other, and *vice versa*.

Private Equilibrium

Suppressing the notation for the shifters, we express the objective function for the risk-neutral firm j that chooses the output q_j and resilience investment I_j to maximize the expected profits, $E[\pi_j]$, under the hazard:

$$(3) \quad \max_{q_j, I_j} E \left[(P^r(Q) - P^f(Q))q_j \right] - C(q_j, I_j),$$

where $Q = \sum_{j=1}^N q_j$ and C is the total cost of processing and retailing that includes the investment in risk reduction. Fixed costs are inconsequential to the short-term production decisions of interest and are, thus, omitted from the setup. Peer firms' choices, $q_{\neq j}$, are taken as given.

Each firm faces a probability of shutdown, ϕ_j . Depending on whether firm j is forced to close, the status-contingent profit changes.⁶ When the firm survives the hazard (denoted by $s_j = 1$) with a chance of $1 - \phi_j$, it earns revenue of $E[P^r(Q) - P^f(Q)|s_j = 1]q_j$ in expectation. When it shuts down (denoted by $s_j = 0$) with a chance of ϕ_j , it produces zero units and earns $E[P^r(Q) - P^f(Q)|s_j = 0] \times 0 = 0$ revenue, but still incurs other processing/retailing costs and the *ex ante* risk management investment. The objective function is rewritten as:

$$(4) \quad \max_{q_j, I_j} E \left[P^r(Q) - P^f(Q)|s_j = 1 \right] (1 - \phi_j)q_j - C(q_j, I_j).$$

To ensure the model's tractability, we let the demand and supply functions be affine hereafter, while keeping the full mathematical complexity of $\phi_j(I_j)$ to capture the entire spectrum of investment interdependence — from complementary to independent to substitutive.⁷ With affine functions, we can write $E[P(Q)]$ as $P(E[Q])$. We rewrite the objective function of the firm:

$$(5) \quad \max_{q_j, I_j} \left(P^r(E[Q|s_j = 1]) - P^f(E[Q|s_j = 1]) \right) (1 - \phi_j)q_j - C(q_j, I_j).$$

The expected market quantity conditional on firm j 's survival, $E[Q|s_j = 1]$, is equal to $\sum_{i=1}^N E[q_i] = \sum_{i \neq j}^N (1 - \phi_i)q_i + q_j$.

We omit the condition of $s_j = 1$ in the following expressions for the ease of presentation. The private Nash equilibrium can be found by taking the first-order conditions (FOCs) with respect to q_j and I_j , respectively.

$$(6) \quad \begin{aligned} & \left(\frac{\partial P^r}{\partial Q} \frac{\partial E[Q]}{\partial q_j} - \frac{\partial P^f}{\partial Q} \frac{\partial E[Q]}{\partial q_j} \right) (1 - \phi_j)q_j + \Delta P(1 - \phi_j) = C_{q_j} \\ & \left(\frac{\partial P^r}{\partial Q} \frac{\partial E[Q]}{\partial I_j} - \frac{\partial P^f}{\partial Q} \frac{\partial E[Q]}{\partial I_j} \right) (1 - \phi_j)q_j - \Delta P q_j \phi_{I_j} = C_{I_j}, \end{aligned}$$

where $\Delta P = P^r(E[Q]) - P^f(E[Q])$, C_{q_j} and C_{I_j} are the partial derivatives of total cost with respect to q_j and I_j , and $\phi_{I_j} = \frac{\partial \phi_j}{\partial I_j} < 0$ is the marginal risk reduction from one additional unit of I_j . The left-hand side (LHS) of the first equation is the marginal revenue of producing one more unit of q_j with $(1 - \phi_j)$ reflecting the expected probability of avoiding shutdown, while the right-hand side (RHS) is the marginal cost of production. The marginal benefit depends on responses of farm and retail prices to market-level outputs. In the second equation, similarly, the LHS is the marginal return to investing an additional unit into risk management, while the RHS reflects the marginal cost of investing. The direct marginal return is the change in revenues realized from reducing ϕ given the output, $-\Delta P q_j \phi_{I_j}$. The indirect marginal return is the change in profit margin via Q given ϕ and q_j .

Plugging in $\frac{\partial E[Q]}{\partial q_j}$ and $\frac{\partial E[Q]}{\partial I_j}$ under symmetry in equilibrium (see derivation in the Mathematical Details appendix), the two equations in system (6) are rewritten as:

$$(7) \quad \begin{aligned} (1 - \phi)P^r \left[1 - \frac{(1 - \phi)\xi + \frac{\phi}{N}}{\eta} \right] &= (1 - \phi)P^f \left[1 + \frac{(1 - \phi)\theta + \frac{\phi}{N}}{\varepsilon} \right] + C_q \\ -P^r q \left[\phi_I - \frac{(1 - \phi)\phi_{II}}{\eta} \right] &= -P^f q \left[\phi_I + \frac{(1 - \phi)\phi_{II}}{\varepsilon} \right] + C_I. \end{aligned}$$

where prices are evaluated at $E[Q] = (1 - \phi)Q + \phi \frac{Q}{N}$, $\phi_I = \frac{\partial \phi}{\partial I}$, $\phi_{\bar{I}} = \frac{\partial \phi}{\partial \bar{I}}$, and $\phi_{II} = \frac{(N-1)\phi_I \xi}{N} + \frac{[(N-2)\xi + 1]\phi_{\bar{I}}}{N}$.⁸

The terms θ and ξ are equal to $\frac{\partial Q}{\partial q} \frac{q}{Q}$ in the first equation of system (7) and are the conjectural elasticities for the buyer and seller power, respectively. When θ (ξ) is equal to 1, the firms run as a cartel (i.e., monopsony/monopoly), obtaining a markdown (markup) over the marginal value of product (marginal cost). When equal to 0, the firm has negligible or no influence on the total market output (i.e., perfect competition). Conjectural elasticities reflecting oligopsonies/oligopolies take on values between 0 and 1 to measure varying degrees of intensity. This model can capture the range of anti-competitive behaviors (e.g., Cournot-Nash or tacit collusion). Parameter $\eta > 0$ is the absolute value of the demand elasticity evaluated at the market equilibrium, and $\varepsilon > 0$ is the farm supply elasticity evaluated at the market equilibrium.

The second equation introduces a third conjectural elasticity, ς , which allows for collusion along the dimension of investing in risk management. The new term captures the fact that firms may form Nash conjectures along a non-price or non-quantity dimension. In this case, $\varsigma = \frac{\partial I_i/I_i}{\partial I_j/I_j} \forall i \neq j$ reflects a conjectural elasticity of risk management. It takes on an interpretation similar to the output conjectural elasticities (ξ and θ) and ranges in $[0, 1]$. Intuitively, it captures the extent to which one firm adjusts its investment given an increase in investment by a peer firm j .

The conjectural elasticity of risk management may take values different from ξ and θ . On the one hand, ς could be smaller because investments by peer companies may be harder to observe, and thus more difficult to collude in or enforce penalty upon, than output by peers. On the other hand, industry-level coordination in risk management may not violate anti-trust policy which focuses on preventing direct control on output. Hence, coordination along non-output mechanisms, like risk technologies or strategies, may be easily formed among firms. For instance, empirical evidence suggests that car manufacturers collude in input technology (Alé-Chilet et al. 2026) and telecommunication firms may collude in product offerings (Bourreau, Sun, and Verboven 2021).

Socially Optimal Equilibrium

A social planner who maximizes the expected social welfare has an objective function different from that of the private firm. Here, we let the social planner determine the risk-reducing investment, and firms choose output given the planner's level of risk management. Within USDA, the Animal and Plant Health Inspection Service (APHIS) mandates biosecurity investments through compulsory vaccination programs, traceability requirements, and movement restrictions during disease outbreaks, approximating the uniform investment standard in our model.⁹ Notably, we do not let the social planner set firm-level outputs, meaning that the social planner would only achieve a second-best solution. We do so for two reasons: (i) in practice, it would require the government to seize control of outputs produced by the private food processing/retailing firms, which seems unlikely, and (ii) if we did, the social planner would simply generate the trivial perfect-competition outcomes in expectation (see the *Alternative Modeling Setup* appendix for details of the first-best solution that could be achieved by the social planner).

The objective function of the social planner is specified as:

$$(8) \quad \max_{I_1 \dots I_N} \sum_{j=1}^N E[\pi_j] + E[CS] + E[PS],$$

where $E[CS]$ is the expected total consumer surplus (CS) and $E[PS]$ is the expected total producer surplus (PS) for the raw input suppliers, or farmers in this case. CS equals $\int_0^Q (P^r(x) - P^r(Q)) dx$ where $Q = Nq$. The final market output varies with the realization of ϕ . Similarly, PS equals $\int_0^Q (P^f(Q) - P^f(x)) dx$.¹⁰

Given the investment solution, I^* , and corresponding $\phi(I^*)$, the firm chooses the output to maximize its expected profits:

$$(9) \quad \max_{q_j} \Delta P(\phi^*)(1 - \phi^*)q_j - C(q_j, I^*),$$

where $\Delta P(\phi^*) = P^r((1 - \phi^*)Q) - P^f((1 - \phi^*)Q)$ because $E[Q] = (1 - \phi^*)Q$ due to symmetry.

Taking the FOC with respect to q_j , we find Q^* as a function of I . Plugging Q^* into the equation (8), we then solve the two-stage game by backward induction.

Analytical Solutions

The affine supply and demand functions are specified as follows:

$$(10) \quad P^f(Q) = b + \beta Q,$$

$$(11) \quad P^r(Q) = a - \alpha Q,$$

CS and PS are hence measured by triangle areas determined by the equilibrium Q . The profits of firms are captured by the rectangular area set by the equilibrium Q .

We let $C = cq_j + I_j q_j$, implying a constant marginal cost of processing/retailing and a per-output investment in risk. The unit cost of investment implies an increasing cost of risk management with the scale of production. Generally, this can be interpreted as production-cost inefficiencies that ensure a more stable production process and output.¹¹ For example, the cost of administering vaccines to each live animal would scale proportionally to output (Hennessy 2007).

We impose an exponential hazard function that has long been used in a variety of contexts to measure the probability of risk exposure or firm closure (Audretsch and Mahmood 1995; Wang and Hennessy 2015):

$$(12) \quad \phi(I_j; I_{k \neq j}) = \lambda e^{-\gamma(I_j + \kappa I_j \bar{I})},$$

where $\lambda \in (0, 1)$ indicates the size of the exogenous hazard. The term $e^{-\gamma(I_j + \kappa I_j \bar{I})}$ captures the endogenous exposure to the hazard, where firm j can lower its exposure through I_j .

This functional form of ϕ captures a critical feature of managing biohazards: the interdependence of risk management among agribusiness firms. That is, one firm's risk management can

also influence other firms' probability of exposure, and *vice versa*. We allow for this possibility captured by the term $\kappa I_j \bar{I}$, where $\bar{I}$ is the average investment by all other firms.

Parameter κ captures the independence (i.e., $\kappa = 0$), the complementarity (e.g., κ is negative and sufficiently large), or the substitutability (e.g., κ is positive and sufficiently small) in investments (see online *Mathematical Details* appendix). For simulations, we set $\kappa \in [-1, 1]$. In this range, $\frac{\partial \phi}{\partial I_j} \leq 0$ and $\frac{\partial^2 \phi}{\partial^2 I_j} \geq 0$, which implies that risk exposure is decreasing in own-firm's risk management investment at a decreasing rate.

Private solutions Due to the symmetry among the firms, we know that the output and investments of the firms would be identical in equilibrium. We rewrite the first equation in system (7):

$$(13) \quad (1 - \Phi)P^r(E[Q]) \left(1 - \frac{X}{\eta}\right) = (1 - \Phi)P^f(E[Q]) \left(1 + \frac{H}{\varepsilon}\right) + c + I,$$

where $\Phi = \lambda e^{-\gamma(I + \kappa I^2)}$ and I is the symmetric investment by a firm, $X = (1 - \Phi)\xi + \frac{\Phi}{N}$, and $H = (1 - \Phi)\theta + \frac{\Phi}{N}$. Term $E[Q]$ equals $(1 - \Phi)Q + \frac{\Phi}{N}Q$. The second equation becomes:

$$(14) \quad -P^r(E[Q]) \left[\Phi_I - \frac{(1 - \Phi)\Phi_{II}}{\eta} \right] = -P^f(E[Q]) \left[\Phi_I + \frac{(1 - \Phi)\Phi_{II}}{\varepsilon} \right] + 1,$$

where $\Phi_I = -\lambda \gamma(1 + \kappa I)e^{-\gamma(I + \kappa I^2)}$, $\Phi_{\bar{I}} = -\lambda \frac{\gamma \kappa I}{N-1} e^{-\gamma(I + \kappa I^2)}$, and $\Phi_{II} = \frac{(N-1)\Phi_I \xi}{N} + \frac{[(N-2)\xi + 1]\Phi_{\bar{I}}}{N}$ (see online *Mathematical Details* appendix for derivation). Because of the nonlinearity in the probability of risk exposure, this system does not permit an analytical solution for I and Q . Therefore, we solve this system numerically in the *Simulation Results* section.

Public solutions Due to symmetry, denote the equilibrium investment and hazard by $\phi^* = \phi(I^*)$. Given ϕ^* and I^* , the FOC for the firm in the second stage is:

$$(15) \quad \begin{aligned} & (1 - \phi^*) \left[1 - \frac{(1 - \phi^*)\xi + \frac{\phi^*}{N}}{\eta} \right] \left\{ a - \alpha \left[(1 - \phi^*) + \frac{\phi^*}{N} \right] Q \right\} \\ & = (1 - \phi^*) \left[1 + \frac{(1 - \phi^*)\theta + \frac{\phi^*}{N}}{\varepsilon} \right] \left\{ b + \beta \left[(1 - \phi^*) + \frac{\phi^*}{N} \right] Q \right\} + c + I^*. \end{aligned}$$

Denoting $x^* = (1 - \phi^*)\xi + \frac{\phi^*}{N}$ and $h^* = (1 - \phi^*)\theta + \frac{\phi^*}{N}$, we obtain the equilibrium output:

$$(16) \quad Q^* = \frac{a \left(1 - \frac{x^*}{\eta} \right) - b \left(1 + \frac{h^*}{\varepsilon} \right) - \frac{c + I^*}{1 - \phi^*}}{\left[(1 - \phi^*) + \frac{\phi^*}{N} \right] \left[\alpha \left(1 - \frac{x^*}{\eta} \right) + \beta \left(1 + \frac{h^*}{\varepsilon} \right) \right]}.$$

Denote the equilibrium output by $Q(I)$, the first stage of the social planner's problem becomes:

$$(17) \quad \max_{I_1 \dots I_N} E[\Pi(Q(I), I)] + \frac{1}{2} E[\alpha Q(I)^2] + \frac{1}{2} E[\beta Q(I)^2],$$

where $E[\Pi(Q(I), I)] = \sum_{j=1}^N E[\pi_j]$ is specified in equation (5). Variable N is the number of firms. Some n firms are hit by a hazard, and we draw n from a Binomial distribution of $n \sim \text{Bin}(N, \phi)$. The expected market output $E[Q]$ equals $Nq(1 - \Phi)$. All else equal, the equilibrium output Q^* increases in a (i.e., demand shifts out) and decreases in b (i.e., supply shifts in). Similarly, one can show that Q^* falls in the seller and buyer power, ξ and θ . While Q^* decreases in I or $\frac{\partial Q^*}{\partial I} < 0$ in general, suggesting that the two actions are strategic substitutes, under a small set of conditions the sign of $\frac{\partial Q^*}{\partial I}$ could be positive. We provide full derivations of such conditions in the online *Mathematical Details* appendix.

Given the linear demand and supply functions, the objective function for the social planner

can be rewritten as follows:

$$(18) \quad \max_I \underbrace{(a-b)(1-\phi)Q(I) - (\alpha+\beta) \left[(1-\phi)^2 + \frac{(1-\phi)\phi}{N} \right] Q(I)^2 - cQ(I) - IQ(I)}_{E[\Pi]} + \underbrace{\frac{\alpha}{2} \left[(1-\phi)^2 + \frac{(1-\phi)\phi}{N} \right] Q(I)^2}_{E[CS]} + \underbrace{\frac{\beta}{2} \left[(1-\phi)^2 + \frac{(1-\phi)\phi}{N} \right] Q(I)^2}_{E[PS]}.$$

Given the symmetry in solution, the FOC is:

$$(19) \quad \underbrace{-\Phi_I Q \left[(a-b) - (\alpha+\beta) \frac{2N-1+2\Phi(N-1)}{2N} Q \right]}_{\text{Direct marginal benefits}} + \underbrace{Q_I \left[(a-b)(1-\Phi) - (\alpha+\beta)(1-\Phi)^2 Q - \frac{(\alpha+\beta)(1-\Phi)\Phi Q}{N} \right]}_{\text{Indirect marginal benefits}} = \underbrace{Q}_{\text{Direct marginal costs}} + \underbrace{Q_I(c+I)}_{\text{Indirect marginal costs}},$$

where $Q_I = \frac{\partial Q}{\partial I}$, $\Phi = \lambda e^{-\gamma(I+\kappa I^2)}$, and $\Phi_I = -\lambda\gamma(1+2\kappa I)e^{-\gamma(I+\kappa I^2)}$. The LHS is the marginal benefit of adding one unit of investment in risk reduction. It includes the marginal increase in $E[\Pi]$ and the marginal increases in $E[CS]$ as well as $E[PS]$. The increase is driven directly by changing the realized output, namely, $-\Phi_I Q$, and indirectly by changing firm output via Q_I given Φ . The RHS is the corresponding marginal cost. Similarly, it includes a direct cost increase of Q and indirect increases via increased output, namely, $(c+I)Q_I$.

Discussion of the Theoretical Solutions

The solutions to the private and public problems retain the conventional properties of an imperfectly competitive market with conjectural variations in output and input procurement, captured by ξ and θ . A key extension of our framework is that firms also strategically interact in risk-management investment, I , with the corresponding conjectural variation captured by ς .

In this subsection, we isolate the economic forces that drive firms' and the social planner's investment decisions, and hence the wedge between private and public risk management. These forces determine both the direction and the magnitude of the investment wedge and the distribution of welfare gains, as the simulations in the *Simulation Results* section demonstrate in an empirically relevant risk-management context.

Direct effect of investment on risk The fundamental incentive for risk-management investment arises from the hazard function itself. Because $\Phi_I < 0$, a marginal increase in I directly lowers the probability of hazard realization, raising expected market output and the value generated in the supply chain.

In the firm's FOC of equation (6), this benefit enters as $-\Delta P \cdot q \cdot \Phi_I > 0$, namely, the expected profit gain from surviving the hazard at a higher rate. In the social planner's FOC of equation (19), the same direct benefit appears through the term $-\Phi_I \cdot Q \cdot \left[(a - b) - (\alpha + \beta) \frac{2N-1+2\Phi \cdot (N-1)}{2N} Q \right]$, which weighs the marginal gain from lower risk by the social value of realized output rather than by the firm's margin alone.

The two problems share the same basic incentive of investing each dollar to reduce risk, but differ in how the gains from risk reduction are valued. This valuation difference contributes to the public-private investment wedge, defined as $\Delta I = I_{gov}^* - I_{firm}^*$. The sign and magnitude of ΔI are shaped by two direct effects and an indirect effect of how I influences Q as illustrated below.

Social welfare effect: The social planner's objective function (18) includes expected consumer surplus, $E[CS]$, expected producer surplus, $E[PS]$, and total firm profits, $E[\Pi]$. Investment that reduces hazard realization raises expected supply, benefiting consumers through lower retail prices and farmers through higher procurement volumes. Private firms do not internalize these welfare gains. For any given Q , the social planner's marginal return to investment, therefore, exceeds the private firm's, implying that $\Delta I > 0$ in isolation. We refer to this component of ΔI as the *social welfare effect*.

The magnitude of this effect depends on the share of total welfare carried by consumers and

producers. Under perfect competition, firm profits approach zero; welfare is composed almost entirely of consumer and producer surplus; and the gap between the planner's and the firm's marginal return is the widest. As quantity market power increases through ξ and θ , firm profits constitute a larger share of total welfare. The planner's valuation of risk reduction becomes closer to the firm's valuation, and the gap narrows.

Investment coordination effect: Each firm's investment raises the industry expected supply and compresses the profit margin ΔP for all firms. If risk management is uncoordinated ($\varsigma=0$), each firm ignores the reduction in peer firms' profits caused by its own investment and tends to over-invest from the aggregate industry's perspective. If firms are able to coordinate in risk management to some extent ($\varsigma>0$), they have an incentive to reduce I relative to the uncoordinated world. This incentive operates similarly to how buyer and seller power lead firms to collectively reduce output to increase profits.

When the social planner imposes a common investment level of I_{gov}^* for all firms, it weighs the benefits of coordinated risk management on industry profits via $E(\Pi)$ in $E(W)$. The *investment coordination effect* works in the opposite direction of the social welfare effect, pushing I_{gov}^* down relative to I_{firm}^* .

The magnitude of this force depends on the share of total welfare captured by firms. Under perfect competition, firm profits are approximately zero, and the social planner places little weight on the effect of coordinated investment on industry profits. As buyer and seller power increase, firm profits constitute a larger share of total welfare, and the coordination effect grows. The coordination effect dissipates when firms already coordinate in investment, namely, as ς rises. When $\varsigma = 1$, the investment coordination effect is absent because firms fully internalize the effect of their investment choices on peer firms' profits.

Indirect effects of investment on output In the social planner's two-stage problem, I^* also depends on how investment affects equilibrium output through $Q_I = \frac{\partial Q^*}{\partial I}$ (see equation (19)). As equilibrium output adjusts with investment, the marginal benefits and costs of adding one dollar in

risk management change accordingly.

Two opposing channels drive Q_I , creating an ambiguous direction of the net indirect effect. The *cost channel*: higher I raises the variable cost $(c + I)$ per unit of output, increasing the effective marginal cost and reducing Q^* . The *reliability channel*: higher I lowers Φ , reducing the effective marginal cost by reducing uncertainty and raising the expected quantity that reliably reaches the market.

When the reliability channel dominates (e.g., when risk is highly responsive to investment and market power is high), $Q_I > 0$, and more risk management boosts output along with its social benefits. When the cost channel dominates, $Q_I < 0$, and firms produce less given the higher per-unit risk management costs. The net sign is parameter-dependent as outlined in the online *Mathematical Details* appendix.

When the cost channel dominates ($Q_I < 0$), imperfectly competitive firms ($\xi/\theta > 0$) partially recoup higher costs by increasing markups (or markdowns). On the other hand, the social planner is disincentivized to increase I in the first stage, because higher costs decrease market supply and hence decrease CS and PS. This tension may lead to firms investing more in I than the social planner.

Net effect of the forces Under the perfectly competitive benchmark ($\xi = \theta = \varsigma = 0$), the social welfare effect is dominant. Thus, private firms tend to under-invest relative to the social planner. As the model deviates from perfect competition, the social welfare margin and the investment coordination margin operate in opposing directions, and neither dominates analytically. The two channels of the indirect output effect further contribute to this ambiguity. We explore the net effect of these forces through Monte-Carlo simulations within a context emblematic of US livestock supply chains in the *Simulation Results* section.¹²

Parameterization

We normalize the risk-free, competitive equilibrium industry-level output, Q^c , to 1.0. The corresponding equilibrium retail price on the national market is $a - \alpha Q^c$ and is also normalized to 1.0. The demand elasticity in this equilibrium, η , hence equals $\frac{1}{\alpha}$, and $a = 1 + \alpha = 1 + \frac{1}{\eta}$.

Similarly, the competitive equilibrium price for the farm is $f = 1 - c$. The farm price is the farm share of the normalized retail revenue from a unit of the final food product under perfect competition. Due to the assumption of fixed proportions and the equalization of units across supply chain stages, the competitive farm output is also 1.0. Thus, $\beta = \frac{f}{\varepsilon}$ and $b = f(1 - \frac{1}{\varepsilon})$, where ε is the elasticity of the farm supply in competitive equilibrium.¹³ The other market parameters are related to buyer and seller power. We start by studying the perfectly competitive case ($\xi = \theta = \varsigma = 0$) in our baseline specification and build up to the imperfectly competitive case.

We ground our simulations in the context of managing disease in the US livestock supply chain and set baseline parameters extracted from prior literature. [Table 1](#) describes these baseline parameter values. In the online *Additional Tables and Figures* appendix, we test the robustness of our findings to alternative parameter choices. In general, the qualitative findings hold across different values of market parameters (see [Figure D2](#)).

Risk Parameters

The remaining parameters relate to the nature of the risk function, $\Phi(I|\lambda, \gamma)$, and, of course, vary by context. We hence do not claim to identify accurate values of these parameters within a specific context, but draw parameter values from a plausible distribution set based on the management of hog diseases and show via simulation how the nature of risk affects the outcomes.

For a given hazard, λ measures the probability of exposure without any investment in risk management, namely, $\phi = \lambda e^0 = \lambda = \phi(0)$. As a baseline, we draw values for λ from a normal distribution $\lambda \sim N(0.25, 0.075)$ such that the hazards are sufficiently large and firms are aware and motivated to protect against them.

Table 1: Baseline Parameter Values

Parameter	Description	Baseline Value	Sim. Range	Justification
<i>Risk Parameters:</i>				
λ	Size of the hazard	$\lambda \sim N(0.25, 0.075)$	$[0, 1]$	Depends on the nature of risk. Baseline values discussed in the online <i>Parameterizing the Risk Technology</i> appendix.
γ	Semi-elasticity of risk investment	$\gamma \sim N(35, 10)$	NA	
κ	Interdependence of I among firms	0	$[-1, 1]$	
<i>Market Parameters:</i>				
ξ/θ	Market power parameter	0	$[0, 0.5]$	(Saitone and Sexton 2017)
α	$\frac{\partial P^r}{\partial Q}$, slope of the demand curve, $\frac{1}{\eta}$	$\eta \sim U(0.46, 0.92)$	NA	(Andreyeva, Long, and Brownell 2010 ; Okrent and Alston 2011)
β	$\frac{\partial P^f}{\partial Q}$, slope of the supply curve, $\frac{f}{\varepsilon}$	$\varepsilon \sim U(0.81, 1.61)$	NA	(Chavas and Cox 1995)
f	Competitive farm share	0.3	NA	Farm share of each retail dollar in a competitive market. See the table note.
N	Number of intermediary firms	4	NA	Set based on a four-firm processing stage (e.g., the US poultry processing), but the value is not consequential for simulation outcomes

Note: The value of competitive farm share out of each retail food dollar is set at 30%, and is set to be a conservative measure calculated from the retail-farm price spread in the meat industry (<https://www.ers.usda.gov/data-products/meat-price-spreads/>). Setting f at different values imposes no qualitative impacts on the simulation results. See online panel (a) of Figure D2.

The value of γ captures the effectiveness of strategies or technologies in reducing risk exposure (e.g., processing safety protocols and animal vaccination). Equivalently, γ can be interpreted as the “semi-elasticity of risk management” (i.e., the percent reduction in risk with respect to a unit change in I with $\kappa = 0$). Mathematically, the term $-\frac{\partial \phi}{\partial I} \frac{1}{\phi}$ is equal to γ if $\kappa = 0$. As stated above, the value of γ is context and strategy specific. Some hazards may be avoided more cheaply (higher values of γ), while others may be costly to influence (low values of γ).

To calibrate the value of γ , we rely on relatively abundant information on diseases and vaccines in the context of the US hog industry. In [Table C1](#), we list the ranges of γ computed in the context of hog diseases (e.g., PEDv). We explain the detailed computation steps based on vaccination trial data in the online *Parameterizing the Risk Technology* appendix. For simulations, we draw values of γ from a normal distribution, $\gamma \sim N(35, 10)$.

Simulation Results

We examine the solution of the model and study the effects of varying the nature of the hazard and market structures via three sets of simulations. For each set of simulations, we vary one parameter at a time, holding all other parameters at their baseline values above. We draw 1,000 values from the distributions specified in [Table 1](#) and generate the mean and standard deviation of the simulation outcomes for each variable of interest.¹⁴

As previously discussed, the difference between the optimal public investment, I_{gov}^* , and the optimal private investment, I_{firm}^* is driven by multiple competing effects. To isolate the role of these effects, we first compare optimal risk management under three discrete scenarios: uncoordinated risk management ($\zeta = 0$), industry-coordinated risk management ($\zeta = 1$), and government-determined risk management.

[Table 2](#) compares the three I^* in a baseline scenario. The difference in investment between row (1) and row (3), $\Delta I = 0.067 - 0.060 = 0.007$, reveals the combined social welfare effect, investment coordination effect, and the indirect cost effects. The difference between row (1) and

Table 2: Risk Management and Welfare Under Discrete Market Structure Scenarios

	Scenario	Parameters	I	π	W
Baseline Risk					
(1)	Uncoordinated I	$\varsigma = 0, \lambda = 0.25$	0.060	0.016	0.776
(2)	Industry-Coordinated I	$\varsigma = 1, \lambda = 0.25$	0	0.080	0.603
(3)	Social Planner Sets I	$\lambda = 0.25$	0.067	0.012	0.777
Higher Risk					
(4)	Uncoordinated I	$\varsigma = 0, \lambda = 0.35$	0.072	0.016	0.765
(5)	Industry-Coordinated I	$\varsigma = 1, \lambda = 0.35$	0.010	0.081	0.577
(6)	Social Planner Sets I	$\lambda = 0.35$	0.079	0.013	0.767

Note: Values reflect the equilibrium solutions and welfare outcomes from the specified scenarios. Unlike the Monte-Carlo simulations, other market parameters are fixed at their median values described in Table 1.

row (2) highlights the coordination effect: Firms invest less under perfect coordination than in the uncoordinated case. The difference between row (2) and row (3), in contrast, highlights the welfare effect by comparing a perfectly coordinated industry against a social planner. Thus, the net ΔI is produced by a positive welfare effect of 0.067 and a coordination effect of -0.060 shaped by the indirect effect. The lower panel compares the same set of scenarios, but with a higher baseline hazard rate ($\lambda = 0.35$), suggesting that the relative size of the two effects depends on the size of the hazard.

The size of both effects is determined by three sets of key parameters, including the magnitude of hazard, the interdependence of risks, and the competitiveness of the market. The simulations proceed in three steps. (1) We hold the market perfectly competitive ($\xi = \theta = \varsigma = 0$) and independent ($\kappa = 0$) and vary the hazard magnitude λ . (2) Still under perfect competition, we vary the interdependence parameter κ . (3) We introduce market power ($\xi/\theta, \varsigma$) and examine its interaction with interdependence.

Magnitude of the Hazard: λ

We start by evaluating how the level of hazard impacts the production and risk management decisions. The parameter $\lambda \in [0, 1]$ is the expected size of the hazard absent risk management. Isolated and small threats to production would take a value close to 0, whereas catastrophic and cascading risks would take values closer to 1.

Risk externality Given baseline conditions listed in [Table 1](#), our first set of results demonstrate that socially optimal risk-reducing investment exceeds what is achieved in the private equilibrium for most levels of hazard, confirming the intuitive argument made in [Baldwin and Freeman \(2022\)](#). Externality in risk investment, though, does not necessarily increase with the magnitude of hazard.

In [Figure 1](#), the solid line represents the mean value of 1,000 simulations at a value of λ with the bands of $1.96 \times$ the standard deviation. Panels (a) and (b) demonstrate that for sufficiently small hazards ($\lambda < 0.03$), the optimal strategy is not to invest in risk management (i.e., the corner solution) in both equilibria. As the magnitude of the hazard increases, the optimal public investment increases, followed by the optimal private investment. For large hazards (e.g., $\lambda = 0.30$), firms would invest 6.9% of each normalized, competitive retail dollar into risk management, while the social planner invests about 7.5%. For more modest hazards ($\lambda = 0.10$), private and public optimal investments are about 2.5% and about 3.2% of the normalized dollar, respectively.¹⁵

Panel (c) plots $\Delta I = I_{gov}^* - I_{firm}^*$ and the corresponding wedge in the risk faced by a firm, $\Delta\Phi = \Phi_{gov}^* - \Phi_{firm}^*$. When the welfare effect dominates under perfect competition, ΔI is non-decreasing in the magnitude of hazard. When ΔI plateaus at a value of $\lambda = 0.12$, the firm would invest about 24% ($\frac{\Delta I}{I_{gov}^*} = \frac{0.008}{0.03}$) less than what the social planner would invest in risk management. Practically, this simulation illustrates that risk externality translates to a public-private investment wedge under perfect competition for $\lambda > 0.02$, and that the wedge is relatively consistent after $\lambda > 0.12$.

Panels (a) and (b) also suggest that λ carries implications for the optimal production levels for firms. For values of $\lambda < 0.03$, the equilibrium quantities, Q_{firm}^* and Q_{gov}^* , stay above the risk-

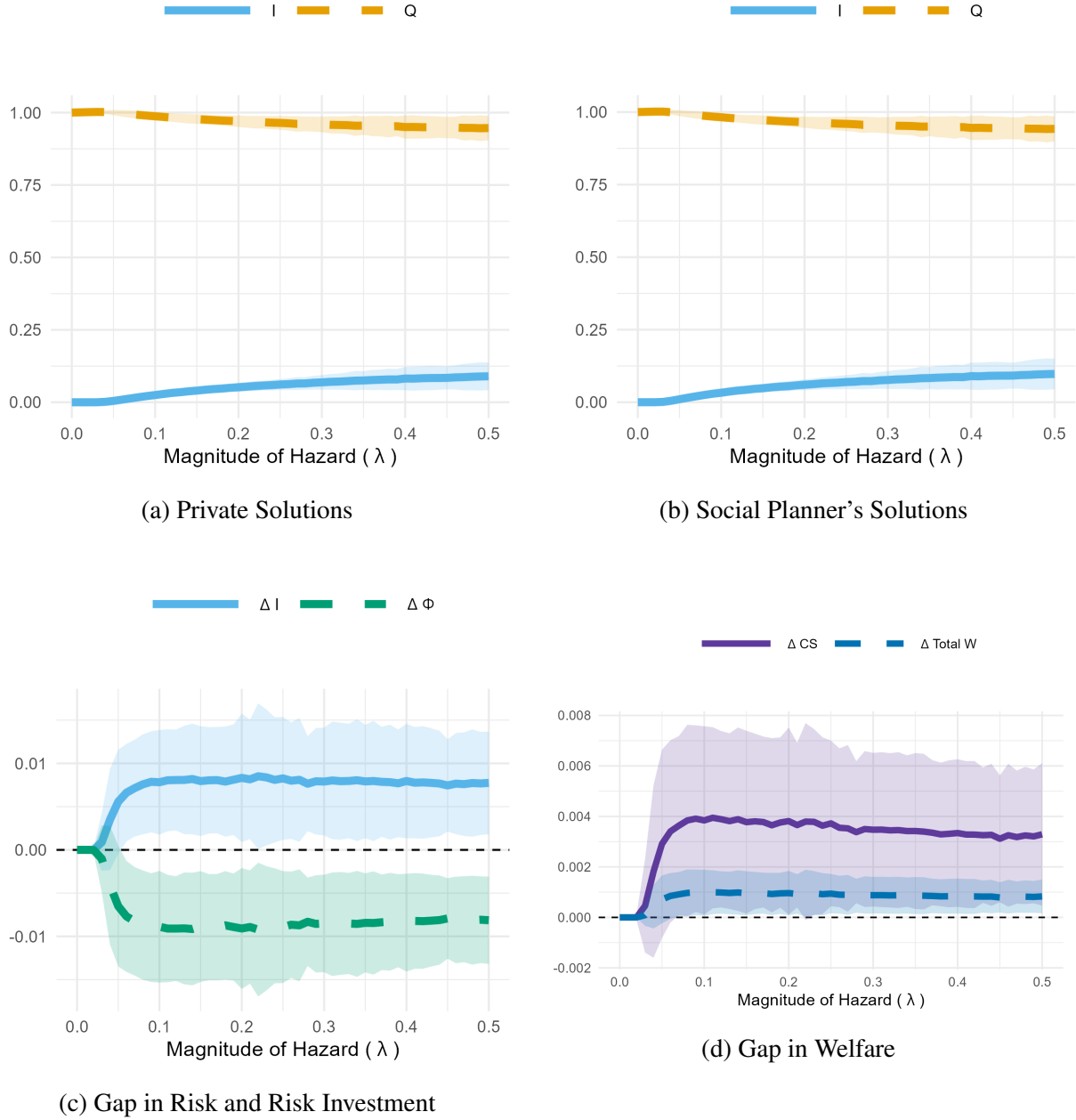


Figure 1: Wedge in risk investment and welfare under competition

Note: Figure displays mean and 1.96-standard-deviation bands from 1,000 numerical simulations with respect to the size of hazard that firms face λ . Δ indicates the difference between the social planner's solutions and the private firm's, I is the investment, Q is the output, Φ is the risk given I , CS is consumer surplus, and W is the social welfare. All welfare terms are measured in expectation. All other parameters are set at baseline values as described in Table 1.

free level, because firms produce slightly more to shield against the potential losses.¹⁶ As risk management expenses move off the corner and increase, firms produce less, following the indirect

cost channel when $Q_I < 0$.

Welfare impacts of risk externality Figure 1d plots the welfare gap between the firm's and the social planner's optimal levels of risk management, namely, $\Delta W = E[W|I_{gov}^*] - E[W|I_{firm}^*]$. At $\lambda = 0.1$, ΔW is worth about 0.1% of the risk-free competitive total welfare. Similarly, ΔCS is computed. At $\lambda = 0.1$, the potential gain in CS from policy intervention is worth $\frac{0.004}{a/2} \times 100 = 0.5\%$ of the competitive, risk-free consumer welfare.

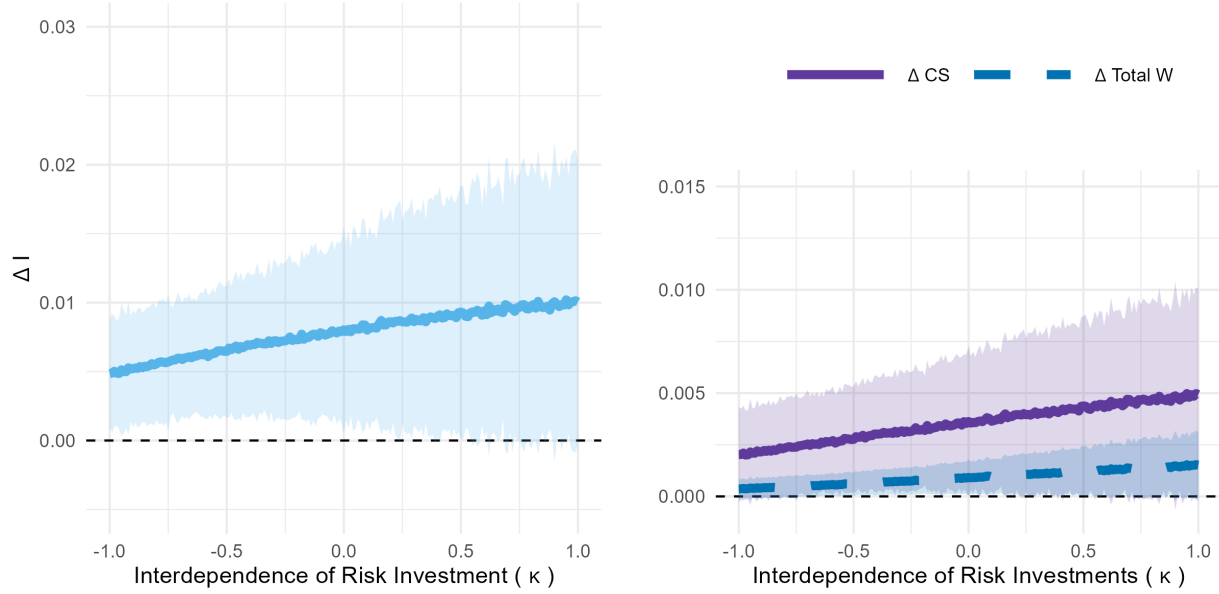
Although not plotted in Figure 1d, ΔPS is proportional to ΔCS (see equation (18)). We omit displaying firm profits ($\Delta \Pi$) from simulation figures because they can be inferred from $\Delta W = \Delta CS + \Delta \Pi + \Delta PS$. By definition of firm profit maximization, $\Delta \Pi \leq 0$ in all scenarios considered (e.g., Figure D3).¹⁷

Interdependence in Risk Management: κ

A firm's risk management of biohazards may be interdependent with other firms. For instance, protective actions taken by one firm may help protect other firms (i.e., creating free-riders). The nature of the dependence may be positive or negative and is captured by κ in the formula of $\phi(I)$. Varying κ from -1 to 1 characterizes various cases in which firm investments are complements, independent, or substitutes.

Figure 2a suggests that ΔI increases in κ . When firm investments are substitutes (i.e., $\kappa \in (0, 1]$), the firm tends to invest even less than the socially optimal level relative to a setting where firm investments are independent because firms tend to free-ride on peers. In contrast, if firm investments are complements (i.e., $\kappa \in [-1, 0)$), ΔI narrows relative to a setting of independent investment because the increased marginal return to individual investment amplified the coordination effect.

Figure 2b displays the corresponding welfare gains from the policy intervention. The welfare improvement from the public intervention increases in κ as the size of the deadweight loss from the externality increases. At a value of $\kappa = 1$, the welfare improvements from intervention



(a) Gap in Risk Investment

(b) Gap in Welfare

Figure 2: Wedge in risk management and welfare under risk dependence

Note: Figure displays mean and 1.96-standard-deviation bands from 1,000 numerical simulations with respect to risk interdependence parameter κ . Δ indicates the difference between the social planner's solutions and the private firm's, I is the investment, CS is consumer surplus, and W is the social welfare. All welfare terms are measured in expectation.

are about 100% higher than the welfare gain at $\kappa = 0$, which is an economically significant difference. This simulation suggests that the returns to public policy intervention tend to be particularly high in contexts where risk management is substitutable among firms.

Conjectural Variations

So far, we have set $\xi = \theta = \varsigma = 0$, meaning that the firm's quantity and investment choices do not influence the peer firms in the market; firms have no market power. However, for a concentrated agrifood processing sector in the US, this assumption is unlikely to hold (Sexton 2013). We now relax the assumption and consider varying values of the quantity and risk conjectures. The conjectures add significant complexity to the public-private wedge in risk investments and potential welfare gains from policy intervention. The interactions of the two types of conjectures are particularly intriguing and discussed carefully in this subsection.

Quantity conjectures: ξ and θ

First, we let ξ and θ vary in a range of $[0, 0.5]$, or from perfect competition to duopoly, while keeping $\varsigma = 0$. Parameters ξ and θ capture the conjecture of the firm on the change in peers' quantity supplied (purchased) given a change in the firm's output (purchase). The larger ξ (θ) is, the greater seller (buyer) power the firm has.

We keep $\xi = \theta$ throughout the simulations.¹⁸ Figure 3a shows that the investment wedge decreases in the conjectural elasticity. At a value $\xi/\theta = 0.2$, ΔI drops to 0, implying that the private, imperfectly competitive firm's incentive to invest in risk management matches that of the social planner because the forces cancel out. Beyond this threshold, the risk externality becomes negative, implying that firms over-invest in risk management from a social welfare perspective.

At first glance, a negative ΔI is perhaps counterintuitive to popular belief. However, this is indicative of forces highlighted in the *Discussion of the Theoretical Solutions* section. As ξ/θ increases, intermediary firms capture greater shares of the total market welfare, and, thus, independently have a greater incentive to protect against downside shocks (i.e., the welfare effect narrows and the coordination effect strengthens). At the same time, higher values of ξ/θ amplify the indirect cost channel, leaving $Q_I \leq 0$. Private firms are incentivized to increase I , because they can offset additional costs by higher markups (or markdowns). By contrast, the socially optimal I remains relatively flat for all values of ξ/θ because the social planner internalizes the dead-weight loss of additional price distortions. The increasing I_{firm}^* and flat I_{gov}^* drive ΔI below zero at approximately $\xi/\theta = 0.2$.

The gap in welfare and who benefits from risk management intervention also varies with respect to the quantity conjectures. For $\xi/\theta < 0.2$, consumers and farmers are the primary beneficiaries of the risk management intervention, improving by as much as 0.5% of their total surplus by moving from the private solution to the public one. However, when $\xi/\theta > 0.2$, the firms are the primary beneficiaries of the government-coordinated risk management. Total welfare still improves as the gains in firm profits outweigh the losses in consumer and producer surplus.¹⁹

The results contribute importantly to public discourse about the role of market power and

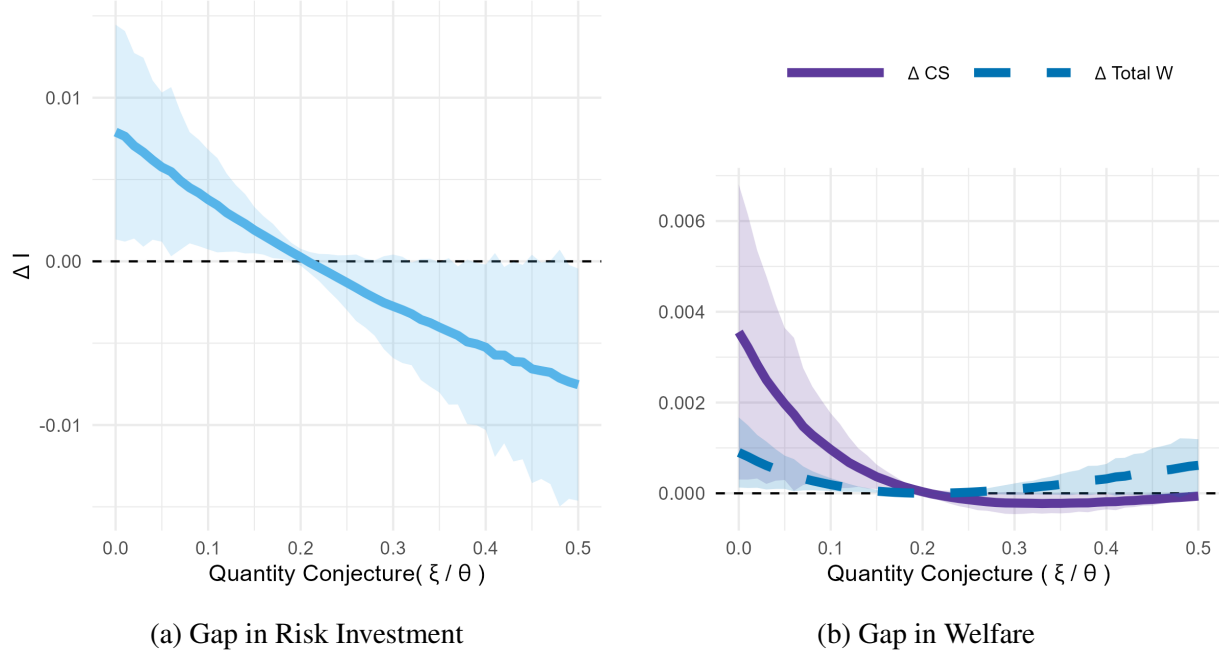


Figure 3: Wedge in risk investment and welfare under market power

Note: Figure displays mean and 1.96-standard-deviation bands from 1,000 numerical simulations with respect to market power parameters ξ and θ with $\varsigma = 0$. Δ indicates the difference between the social planner's solutions and the private firm's, I is the investment, CS is consumer surplus, and W is the social welfare. All welfare terms are measured in expectation. All other parameters are set at baseline values described in [Table 1](#).

supply chain resilience. The popularly held belief, including the official stance from the US Department of Justice, is that greater competition leads to more resilient markets.²⁰ We offer evidence to the contrary and show that firms with more market power may actually contribute positively to resilience because of their greater incentive to protect profits than in the perfectly competitive case. The baseline total welfare still decreases as firms extract surplus from input suppliers and their customers, but at least after a certain threshold, more highly concentrated markets would provide more protection from hazards than would the social planner's solutions. This opposes the findings by [Hadachek, Ma, and Sexton \(2024\)](#) who assume exogenous risks and show that market power harms the resilience of agrifood supply chains. Endogenous risk management by firms may be the underlying theoretical reason why empirical studies on supply chain resilience, such as [Gnutzmann, Kowalewski, and Śpiewanowski \(2020\)](#) and [Richards, Polyviou, and Ubilava \(2025\)](#), find that more concentrated markets tend to be more resilient to shocks.

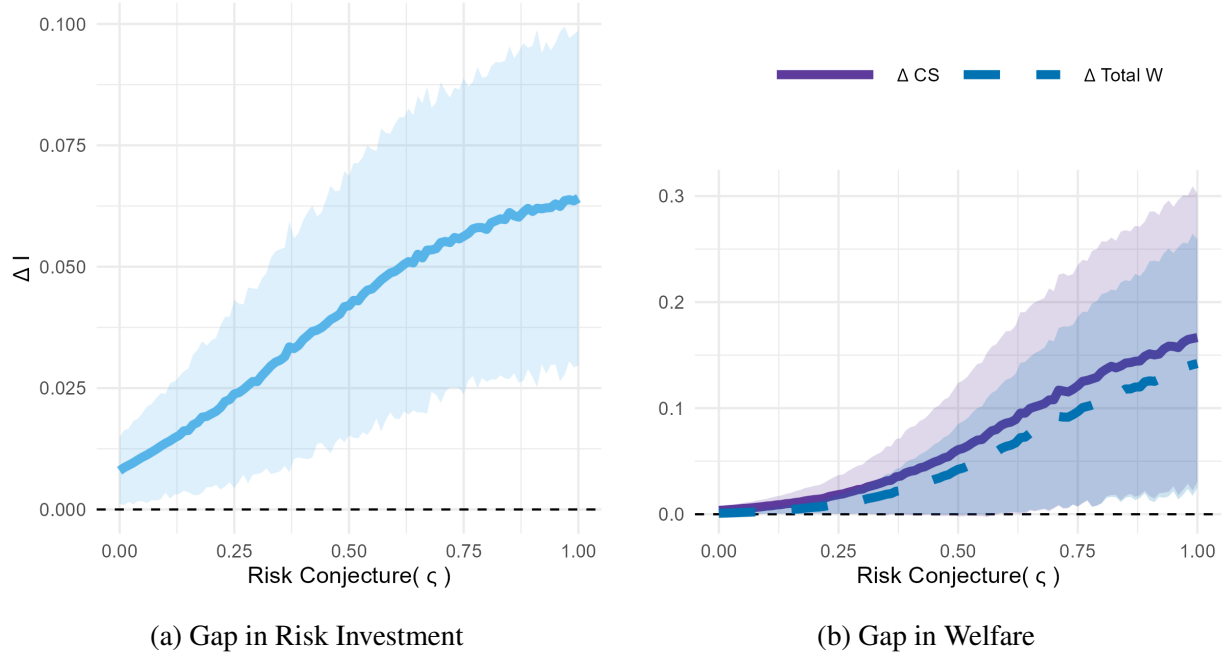


Figure 4: Wedge in risk investment and welfare under risk conjecture

Note: Figure displays mean and 1.96-standard-deviation bands from 1,000 numerical simulations with respect to risk conjecture parameter ζ with ξ and θ equal zero. Δ indicates the difference between the social planner's solutions and the private firm's, I is the investment, CS is consumer surplus, and W is the social welfare. All welfare terms are measured in expectation. All other parameters are set at baseline values described in Table 1.

Investment conjecture: ζ

Coordination among intermediary firms may also enter through the channel of risk management. The degree of risk management coordination is captured by ζ and reflects the firm's conjecture on the change in peers' investment given a change in its own investment. The larger ζ , the greater control the firm has over market-level risk exposure through its own choice of I_j . The government imposes the same I across firms and effectively ensures $\zeta = 1$ among firms.

Figure 4a shows that as ζ increases given $\xi = \theta = 0$, ΔI widens as firms invest less relative to the social optimum because they can better coordinate risk-reducing investment in the same way that firms better coordinate in output with larger ξ and θ . The welfare gains for farmers and consumers from closing ΔI also increase, while firm profits fall.

When ζ is large, the coordination effect shrinks, approaching the comparison between row (2) and row (3) in Table 2. The welfare effect becomes dominant, and magnitudes of the risk

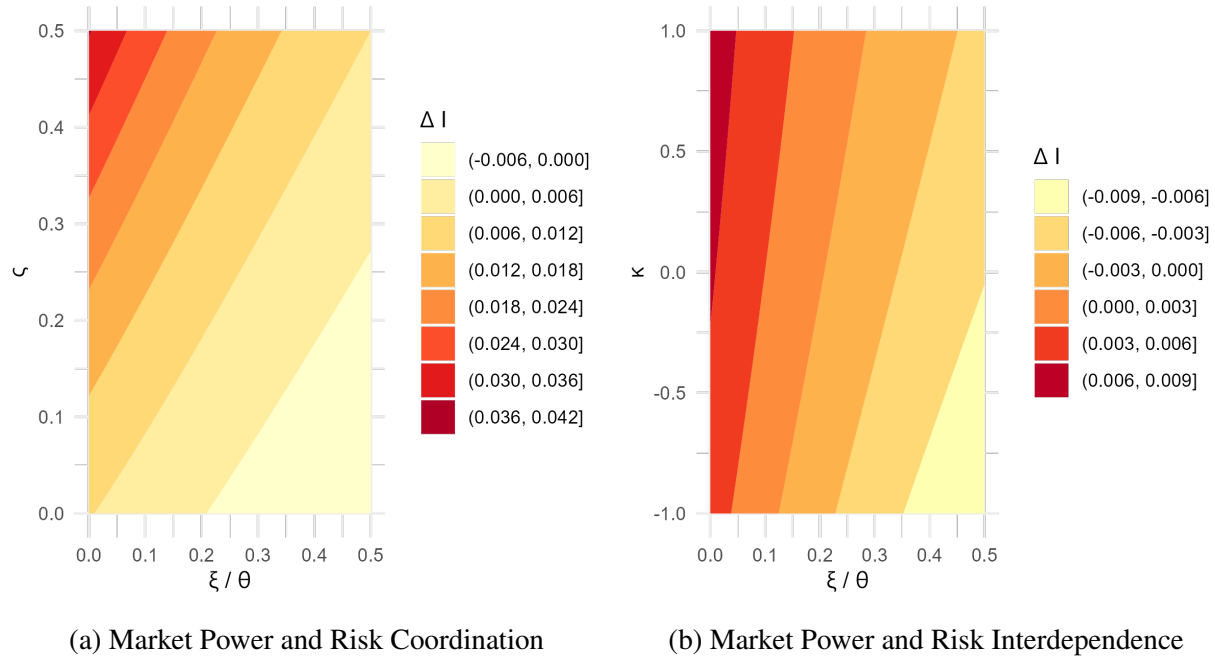


Figure 5: Interaction of risk interdependence and market structure

Note: Figure displays the wedge in risk investment ΔI plotted over differing values of conventional buyer/seller power ξ/θ , the risk conjecture ζ , and risk interdependence κ .

externality and welfare wedges are larger relative to the simulations where ζ is small. For instance, when $\zeta = 0.4$, firms under-invest in risk management by 54% of the social optimum. In this case, the total deadweight loss from letting firms dictate risk management is about 3.6% of total welfare achieved by the social planner at the same value of ζ . Consumers would stand to benefit the most from government intervention, gaining as much as 7.0% of the baseline CS.

Interactions of Risk and Market Characteristics

When varied independently, simulations so far have shown that the quantity conjectures (ξ/θ) and the investment conjecture (ζ) draw different implications for policy justification and welfare impacts. When varied jointly, the implications of conjectures on risk externality and welfare are particularly complex as the relative magnitudes of the social welfare effect, the investment coordination effect, and the indirect cost channel are all simultaneously changing.

Figure 5a displays the interaction of quantity and risk conjectures in a contour plot. When ζ

Table 3: Summary of Simulation Results

Varying Parameter	Risk Parameters	Market Parameters	Effects on Risk Externality (ΔI)	Effects on Welfare Gap (ΔCS)
<i>Risk Parameters:</i>				
$\lambda \uparrow$	$\kappa = 0$	$\xi = \theta = \varsigma = 0$	ΔI is zero initially and then increases before leveling off	ΔCS is zero initially and then increases before leveling off
$\kappa \uparrow$	$\lambda \sim N(\cdot)$	$\xi = \theta = \varsigma = 0$	ΔI increases as κ goes from -1 to 1	ΔCS increases as κ goes from -1 to 1
<i>Market Parameters:</i>				
$\xi/\theta \uparrow$	$\kappa = 0$	$\varsigma = 0$	ΔI decreases, initially positive before falling negative	ΔCS decreases from being positive to negative
$\varsigma \uparrow$	$\kappa = 0$	$\xi = \theta = 0$	ΔI increases	ΔCS increases
<i>Interaction of Parameters:</i>				
ξ/θ and ς	$\kappa = 0$	N/A	With a larger ς , ΔI decreases at a faster rate as ξ/θ rises	With a larger ς , ΔCS decreases more rapidly as ξ/θ increases
ξ/θ and κ	$\lambda = 0.25$	$\varsigma = 0$	With a larger ξ/θ , ΔI increases at a faster rate as investments become more substitutable	With a larger ξ/θ , ΔCS increases faster as κ increases

Note: Authors' creation based on the simulation outcomes presented in this section. See Table 1 for definitions of the parameters. The risk externality, ΔI , equals the optimal investment in risk management by the government minus that by the firm. The potential change in consumer surplus from government intervention, ΔCS , is the consumer surplus obtained in the socially optimal equilibrium minus that in the private equilibrium.

is small, both the social welfare and the investment coordination effects are strong. The externality in investment falls from positive to negative as the conventional ξ/θ rises. In contrast, with a larger ς , the externality falls faster as ξ/θ rises (i.e., $\frac{\partial^2 \Delta I}{\partial \xi \partial \varsigma} < 0$), but stays positive because the coordination effect is trivial and the social welfare effect dominates.

Furthermore, Figure 5b shows the interaction between quantity conjecture and interdependence in risk management, with ξ/θ varying along the horizontal axis and $\varsigma = 0$. Given κ , increasing market concentration reduces the externality in risk management. Like in Figure 3a, the externality turns negative, meaning that the firms invest more than the social planner does, when ξ/θ is large. The externality widens if the firm investments are substitutes instead of complements, and it widens at a faster rate with larger ξ/θ (i.e., $\frac{\partial^2 \Delta I}{\partial \kappa \partial \xi} > 0$).

Summary of simulations Key simulation results in this section are summarized in [Table 3](#). We argue that, when examining supply chain resilience, it is critical to account for the endogeneity of risk management in agrifood supply chains. The endogenous risk management is driven by the nature of imperfect competition — whether in quantity or in risk management practices — and the nature of investment interdependence — either investments by firms are complements or substitutes. Understanding the complex interactions of risk and market conditions plays a vital role in justifying policies aimed at increasing supply chain resilience.

Concluding Remarks

Increasingly frequent disruptions in agrifood supply chains have inspired various policy interventions in the United States and beyond. Interventions may be justified because the socially and privately optimal levels of supply chain resilience differ. A public-private wedge in risk management exists mainly because the firm does not incorporate the impacts of supply chain disruptions, like food insecurity, on consumer or producer welfare, nor does it fully incorporate how its risk management affects peer firms.

We propose a model to characterize the private and public problems of investing in risk management, applied to the context of biohazards in the US agrifood sector. Both the private firm and social planner weigh the costs of risk management against the benefits of more stable market outcomes. The model incorporates inter-firm complementarity and substitution in risk management, a key feature of agricultural production and a novel feature relative to existing models on supply chain resilience. The model also allows for various degrees of market power in processing/retailing to reflect the reality of US agricultural markets ([Sexton 2013](#)).

Simulations demonstrate the potential risk externality and welfare gains from policy intervention under a number of plausible market and hazard contexts. Contrary to popular discourse, a processing/retailing stage with greater buyer and seller power reduces and may even flip the risk externality as oligopolistic firms internalize a larger share of the potential losses from disruptions,

but do not coordinate well in risk management. If firms coordinate closely in risk management, there is considerable justification for policy intervention that would provide large social gains. The gains depend, critically, on inter-firm complementarity or substitution in risk management.

Our model focuses purely on externalities associated with lost output to the market from the firm's risk management decisions. Specific risks and mitigation strategies may involve additional externalities on consumers that depend on the nature of the risk and the mitigation strategy itself. For example, the loss of consumer confidence in a product following adverse events or public health impacts tied to outbreaks or food recalls impacts consumers beyond just the costs of risk management. On the other hand, risk management tools themselves, like vaccinations or pesticides, may generate negative externalities to the environment that are not captured by our model. These factors warrant careful empirical and policy attention in determining optimal risk management strategies and should be considered alongside the risk externalities characterized in this study.

In summary, the nature of coordination among firms in an imperfectly competitive market determines the direction and magnitude of risk externality in a nonlinear way. The role of interdependence in risk management is also critical. All else the same, the more substitutable and cooperative risk investments are among firms, the higher potential welfare gains are from policy intervention. Future empirical work is needed to estimate the degree of risk coordination (ζ) and interdependence (κ) by adapting the conduct parameter methods used in the empirical industrial organization literature, combined with cost structure data on agricultural processing firms ([Schroeter 1988](#)). Extensions of the theory should focus on dynamic behaviors of the firm for managing risk, including fixed-cost investments and entry/exit behavior amid risk.

Footnotes

¹We consider *ex ante* risk management strategies. Firms may also invest in insurance that provides *ex post* revenue protection in the event of a hazard, but we focus on protective strategies that shield against potential inventory or production losses.

²A growing body of theoretical work endogenizes risk strategies by firms and governments via endogenous supply network formation and measures the welfare implications of disruptions within the network (Capponi, Du, and Stiglitz 2024; Grossman, Helpman, and Sabal 2024; Kopytov et al. 2024). Some empirical work has documented that firms adapt to climate risks or natural disasters by adjusting the composition of suppliers (Castro-Vincenzi et al. 2024; Grover and Kahn 2024). These studies, though, do not incorporate the characteristics of biohazards because they are tailored to fit non-agrifood industries.

³One such example is the Secure Pork Supply organization (<https://www.securepork.org/>) that is a coalition between the swine industry, researchers, and government agencies to enhance disease preparedness.

⁴In the livestock sector, details of contractual obligations, including requirements for animal health and handling, can be found through USDA Agricultural Marketing Service contract library. For example, swine contracts can be found at <https://www.ams.usda.gov/rules-regulations/packers-and-stockyards-act/regulated-entities/swine-contract-library>. In the contracts, hog farmers are obligated to meet a set of sanitation, immunization, and other biosecurity protocols set by slaughter firms.

⁵Many major food products are perishable and have limited shelf life, limiting the use of stored products in the case of shuttered production. For non-perishable agricultural products like grain, there is a long strand of literature on storage and commodity price-stabilization policies (Wright 1979; Miranda and Helmberger 1988; Gouel 2013).

⁶Rather than considering a few firms that fully shut down, an alternative and mathematically equivalent interpretation is that all firms experience a partial shutdown from the same shock and lose a common share of their output.

⁷By assuming affine functions, we effectively focus on the first-order Taylor approximation of demand and supply. This simplification is widely used in the literature (Gardner 1975; Cakir and Balagtas 2012; Moschini, Lapan, and Kim 2017; Pi and Vukina 2025). If we allow nonlinearity in demand or supply, the expectation of prices would contain the variance of Q and the second derivative of this deterministic function with respect to Q . Both are tedious mathematical terms that significantly complicate the first-order conditions of the firm and government problems and sacrifice clear economic interpretation of comparative statics. We acknowledge, though, that this simplification implies varying elasticity in simulations. We have examined the sensitivity of the results by assuming constant elasticity instead and confirm the qualitative insights derived from the baseline setup via additional simulations.

⁸Notably, the first equation collapses to the traditional FOOM condition in the risk-free world (i.e., $\phi = 0$). See the FOC in the traditional FOOM setup in, for instance, Hadachek, Ma, and Sexton (2024).

⁹Less stringent approaches include conditional subsidies or tax incentives tied to meeting target criteria. We note that policy implementation in real-world settings is necessarily imperfect and context-specific, whereas our model provides a stylized benchmark for policies that maximize economic efficiency. Moreover, equal investment arises in our study as a consequence of symmetry assumptions; policies may instead impose minimum standards or allow for heterogeneous investment across firms. Our focus is on the role of market structure in shaping risk management

incentives under interdependent risks, and we hence abstract from firm heterogeneity and detailed policy design.

¹⁰With affine demand and supply functions, CS is a triangle above P^r , PS is a triangle below P^f , and profits form a rectangle between the triangles.

¹¹Some risk management strategies may more closely resemble fixed cost or capital investments (e.g., added infrastructure for buffer capacity, mechanical systems, or entry/exit decisions). However, appropriately evaluating the payoffs of such forms of risk management would require a multiple-period model that accounts for the life of irreversible investment. Extending our analysis in this direction would require a dynamic investment model, in the spirit of the strategic investment model by Fudenberg and Tirole (1984) or the real-options approach to irreversible investment under uncertainty developed by Dixit and Pindyck (1994). We focus, instead, on the variable costs of risk management in this study.

¹²In the online *Mathematical Details* appendix, we further demonstrate that the difference between private and public investments is driven by differences in their profits out of certainty-equivalent outputs (i.e., a deterministic output equal to its expected value) and penalties on profits due to uncertainty in realizing the outputs. Their certainty-equivalent profits differ only because the social planner incorporates CS and PS beyond firm profits, which implies a higher *I ceteris paribus* (i.e., the social welfare effect). Their penalty terms share a direct effect of reducing ϕ , but differ because I has an ambiguous effect on the total output $Q(I)$ chosen by firms under social planning as expression (A14) specifies. The indirect effect has an ambiguous sign because investment simultaneously raises production costs and reduces failure risk, and the equilibrium output depends on which of these opposing forces is stronger. As a result, the sign of the difference between private and public investments could go either way.

¹³Importantly, the value of elasticity varies along the linear demand and supply curves by construction. Thus, we only assign $\eta = 0.7$ and $\varepsilon = 1.0$ as the means in the corresponding distributions in the competitive equilibrium and let the elasticity be determined by the corresponding price and output solutions in the simulation. The formulas for prices and output are rewritten in the online *Mathematical Details* appendix.

¹⁴Numerical results are solved via the Newton-Raphson method. We list starting values, convergence rates, and corner solution rates in online appendix Table D1. Under lower-risk scenarios, convergence rates are lower due to a higher share of corner solutions for I .

¹⁵The Risk Management Society's Benchmark Survey, which provides the most detailed record of risk management spending to our knowledge, reports that companies spend about \$9.95 for every \$1,000 of revenue, namely, 1.0% (Wilkinson 2019). This aligns well with the simulation outcomes at $\lambda = 0.07$.

¹⁶This stands in contrast to the findings of Sandmo (1971) which show that price uncertainty causes risk-averse managers to produce less. Our finding here differs because the operational risk enters through the endogenous quantity variable, so firms can directly affect expected revenue by increasing quantity. In Sandmo's model, however, the uncertainty is exogenous.

¹⁷Although this is the perfectly competitive case, firms may still earn a positive profit. When a shock is realized, the total output falls and the equilibrium retail price increases, creating the possibility of positive profits in both the private equilibrium and the social planner's solutions for $\lambda > 0$.

¹⁸Anti-trust policy may focus on just buyer or seller power. However, in the model, the symmetry of linear demand and supply functions ensures that the roles of the two output conjectures are qualitatively indifferent. [Figure D1](#) independently varies buyer and seller power parameters in simulation and confirms that they influence ΔI in qualitatively similar ways.

¹⁹An alternative social planner objective function that more highly values the economic rents of consumers and producers over the imperfectly competitive firms would yield different simulation results in this exercise. This, however, would require an entirely different setup and conditions than the objective function outlined above.

²⁰In [Antitrust Division, US Department of Justice \(2024\)](#), it is stated, "... the Antitrust Division's view is that supply chains are likely to be more resilient when competition is robust and consumers at every level of a supply chain are not dependent on a monopoly supplier or small oligopoly of suppliers."

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AJAE Appendix for
Risk Management Externalities in Agrifood Supply
Chains

Jeffrey Hadachek
Meilin Ma

August 7, 2026

Note: The material contained herein is supplementary to the article named in the title and
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Mathematical Details

This appendix gives detailed mathematical derivations of the key equations and other expressions in the article.

Expectation of Variables

When variable X follows the Binomial distribution of $(N, 1 - \phi)$ among firms, the probability mass function (pmf) of $X = n$ is $C_N^n (1 - \phi)^n \phi^{N-n}$. The sum of the probabilities, $\sum_{n=0}^N C_N^n (1 - \phi)^n \phi^{N-n} = 1$ by definition.

The expectation of X , $E[X]$, equals

$$\sum_{n=0}^N n C_N^n (1 - \phi)^n \phi^{N-n}.$$

Dropping the $n = 0$ term as it equals zero, we have $E[X] = \sum_{n=1}^N n C_N^n (1 - \phi)^n \phi^{N-n}$. Given the binomial-coefficient identity that $n C_N^n = N C_{N-1}^{n-1}$, we have the following equations.

$$\begin{aligned} E[X] &= \sum_{n=1}^N N C_{N-1}^{n-1} (1 - \phi)^n \phi^{N-n} \\ &= N(1 - \phi) \sum_{n=1}^N C_{N-1}^{n-1} (1 - \phi)^{n-1} \phi^{(N-1)-(n-1)} \\ &= N(1 - \phi) \sum_{m=0}^M C_M^m (1 - \phi)^m \phi^{M-m} \\ &= N(1 - \phi), \end{aligned} \tag{A1}$$

where $m = n - 1$ and $M = N - 1$ and $\sum_{m=0}^M C_M^m (1 - \phi)^m \phi^{M-m} = 1$ by the property of Binomial distribution.

Similarly, we can derive the expectation of X^2 using the same technique.

$$\begin{aligned}
 (A2) \quad E[X^2] &= \sum_{n=0}^N n^2 C_N^n (1-\phi)^n \phi^{N-n} \\
 &= \sum_{n=1}^N n C_N^n (1-\phi)^n \phi^{N-n} + \sum_{n=1}^N n(n-1) C_N^n (1-\phi)^n \phi^{N-n},
 \end{aligned}$$

where the first term is computed in equation (A1). We focus on finding the value of the second term.

Using the binomial-coefficient identity again, the second term can be rewritten as

$$\begin{aligned}
 (A3) \quad & \sum_{n=2}^N n(n-1) C_N^n (1-\phi)^n \phi^{N-n} \\
 &= \sum_{n=2}^N (n-1) N C_{N-1}^{n-1} (1-\phi)^n \phi^{N-n} \\
 &= N(1-\phi) \sum_{n=2}^N (n-1) C_{N-1}^{n-1} (1-\phi)^{n-1} \phi^{(N-1)-(n-1)} \\
 &= N(1-\phi) \sum_{n=2}^N (N-1) (1-\phi) C_{N-2}^{n-2} (1-\phi)^{n-2} \phi^{(N-2)-(n-2)} \\
 &= N(N-1) (1-\phi)^2 \sum_{l=0}^L C_L^l (1-\phi)^l \phi^{L-l} \\
 &= N(N-1) (1-\phi)^2,
 \end{aligned}$$

where $l = n - 2$ and $L = N - 2$ and $\sum_{l=0}^L C_L^l (1-\phi)^l \phi^{L-l} = 1$ by the property of Binomial distribution.

Adding up the values of the first and second terms in equation (A2), we have

$$\begin{aligned}
 (A4) \quad E[X^2] &= E[X]^2 + Var[X] = N(1-\phi) + N(N-1)(1-\phi)^2 \\
 &= N^2(1-\phi)^2 + N(1-\phi)\phi \\
 &= N^2 \left[(1-\phi)^2 + \frac{(1-\phi)\phi}{N} \right].
 \end{aligned}$$

Here, $N^2 \frac{(1-\phi)\phi}{N}$ is the aggregate variance of X due to individual probabilities of $1 - \phi$.

Partial Derivatives

We give a detailed derivation for a few complicated partial derivatives the *Analytical Solutions* section below.

First order conditions In the manuscript, $E[Q] = \sum_{i \neq j}^N (1 - \phi_i) q_i + q_j$. Thus, we have

$$\frac{\partial E[Q]}{\partial q_j} = \phi_j + (1 - \phi_j) + \sum_{i \neq j}^N (1 - \phi_i) \frac{\partial q_i}{\partial q_j}.$$

Due to symmetry, the equilibrium q_i is equal across firms, and so is the equilibrium ϕ_i . Thus, we have $\frac{\partial E[Q]}{\partial q_j} = \phi + (1 - \phi) \frac{\partial Q}{\partial q}$. The term $\frac{\partial Q}{\partial q}$ captures the conjecture that a firm has regarding how its competitors react to its output decision. No single firm affects market output under perfect competition, and $\frac{\partial Q}{\partial q}$ is hence zero. The term is positive for imperfect competition cases. The larger the term, the closer the market is to a monopoly scenario.

The term $\frac{\partial E[Q]}{\partial I_j}$ equals

$$-\sum_{i \neq j}^N \frac{\partial \phi_i}{\partial I_j} q_i = -\sum_{i \neq j}^N \left(\frac{\partial \phi_i}{\partial I_i} \frac{\partial I_i}{\partial I_j} + \frac{\partial \phi_i}{\partial \bar{I}} \frac{\partial \bar{I}}{\partial I_j} \right) q_i.$$

Let $\frac{\partial \phi_i}{\partial I_i} \frac{\partial I_i}{\partial I_j} = \frac{\partial \phi_i}{\partial I_i} \varsigma$, with ς reflecting the common conjecture in risk management and ranging in $[0, 1]$. The term $\frac{\partial E[Q]}{\partial I_j}$ hence equals $-\sum_{i \neq j}^N \left(\frac{\partial \phi_i}{\partial I_i} \varsigma + \frac{\partial \phi_i}{\partial \bar{I}} \frac{\partial \bar{I}}{\partial I_j} \right) q_i$. For symmetry and because $\bar{I} = \frac{\sum_{i \neq j} I_i}{N-1}$ for firm j , $\frac{\partial E[Q]}{\partial I_j}$ generates

$$-(N-1) \left(\frac{\partial \phi}{\partial \bar{I}} \varsigma + \frac{\partial \phi}{\partial \bar{I}} \frac{(N-2)\varsigma + 1}{N-1} \right) q = -\{(N-1)\phi_I \varsigma + [(N-2)\varsigma + 1]\phi_{\bar{I}}\} q.$$

For easier notation, denote $(N-1)\phi_I \varsigma + [(N-2)\varsigma + 1]\phi_{\bar{I}}$ by L . Given the symmetry of

firms, the FOCs in system (5) with respect to q_j and I_j , respectively, are rewritten below.

$$(A5) \quad \begin{aligned} \frac{\partial P^r}{\partial Q} \left[\phi + (1-\phi) \frac{\partial Q}{\partial q} \right] (1-\phi)q - \frac{\partial P^f}{\partial Q} \left[\phi + (1-\phi) \frac{\partial Q}{\partial q} \right] (1-\phi)q + \Delta P(1-\phi) &= C_q \\ - \left(\frac{\partial P^r}{\partial Q} Lq - \frac{\partial P^f}{\partial Q} Lq \right) (1-\phi)q - \Delta P q \phi_I &= C_I \end{aligned}$$

In each equation above, we factor out the price evaluated at $E[Q]$ and write terms in the form of elasticity. The system of equations become

$$(A6) \quad \begin{aligned} &(1-\phi)P^r \left[1 + \frac{\partial P^r/P^r}{\partial Q/Q} \phi \frac{q}{Q} + \frac{\partial P^r/P^r}{\partial Q/Q} (1-\phi) \frac{\partial Q/Q}{\partial q/q} \right] \\ &= (1-\phi)P^f \left[1 + \frac{\partial P^f/P^f}{\partial Q/Q} \phi \frac{q}{Q} + \frac{\partial P^f/P^f}{\partial Q/Q} (1-\phi) \frac{\partial Q/Q}{\partial q/q} \right] + C_q, \end{aligned}$$

where $\frac{\partial P^r/P^r}{\partial Q/Q} = -\frac{1}{\eta}$ and $\frac{\partial P^f/P^f}{\partial Q/Q} = \frac{1}{\varepsilon}$. The term $\frac{q}{Q} = \frac{1}{N}$ in equilibrium.

$$(A7) \quad \begin{aligned} &-P^r q \left[\phi_I + \frac{\partial P^r/P^r}{\partial Q/Q} (1-\phi) L \frac{q}{Q} \right] \\ &= -P^f q \left[\phi_I + \frac{\partial P^f/P^f}{\partial Q/Q} (1-\phi) L \frac{q}{Q} \right] + C_I, \end{aligned}$$

where $\phi_I = \frac{\partial \phi}{\partial I}$ and $\phi_{\bar{I}} = \frac{\partial \phi}{\partial \bar{I}}$.

Investment interdependence For the parameterized hazard in the *Analytical Solutions* section and with $\zeta = 0$, the second-order cross derivative of ϕ is $\frac{\partial^2 \phi}{\partial I_j \partial \bar{I}}$. Due to symmetry in equilibrium of I , the derivative equals:

$$(A8) \quad \begin{aligned} &\frac{\partial -\lambda \gamma (1 + \kappa \bar{I}) e^{-\gamma(I_j + \kappa I_j \bar{I})}}{\partial \bar{I}} \\ &= -\lambda \gamma \kappa (1 - \gamma I - \gamma \kappa I^2) e^{-\gamma(I + \kappa I^2)} \end{aligned}$$

The sign of this derivative is determined by the term, $\kappa(1 - \gamma I - \gamma \kappa I^2)$. When κ is positive and larger (smaller) than $\frac{1-\gamma I}{\gamma I^2}$, the derivative is positive (negative). Alternatively, when κ is

negative and smaller (larger) than $\frac{1-\gamma'}{\gamma'^2}$, the derivative is positive (negative).

Certainty Equivalent

Following the analytical solutions, we first define the certainty-equivalent quantity, q^{CE} , as

$$q^{CE} = (1 - \phi)q.$$

Given linearized demand and supply, we can express equation (5) of firm expected profits using q^{CE} .

$$\begin{aligned}
 E[\pi_j] &= E \left[P^r(Q) - P^f(Q) | s_j = 1 \right] (1 - \phi_j)q_j - (c + I_j)q_j \\
 &= [a - \alpha E[Q] - (b + \beta E[Q])] (1 - \phi_j)q_j - (c + I_j)q_j \\
 (A9) \quad &= \left[(a - b) - (\alpha + \beta) \left(\sum_i (1 - \phi_i)q_i + \frac{\phi_j(1 - \phi_j)q_j}{1 - \phi_j} \right) \right] (1 - \phi_j)q_j - (c + I_j) \frac{(1 - \phi_j)q_j}{1 - \phi_j} \\
 &= \underbrace{\left[(a - b) - (\alpha + \beta)Q^{CE} \right] q_j^{CE} - \frac{c + I_j}{1 - \phi_j} q_j^{CE}}_{CE} - \underbrace{(\alpha + \beta) \frac{\phi_j}{1 - \phi_j} (q_j^{CE})^2}_{Penalty}.
 \end{aligned}$$

The terms of $\left[(a - b) - (\alpha + \beta)Q^{CE} \right] q_j^{CE} - \frac{c + I_j}{1 - \phi_j} q_j^{CE}$ reflect the expected profits if q_j^{CE} is certain, or π_j^{CE} . The term $-(\alpha + \beta) \frac{\phi_j}{1 - \phi_j} (q_j^{CE})^2$ captures a quadratic penalty due to operational uncertainty facing hazard ϕ .

As a result, the FOC with respect to q_j^{CE} with a hazard has an additional term of $-2(\alpha + \beta) \frac{\phi_j}{1 - \phi_j} q_j^{CE}$ compared to the certainty-equivalent problem. The FOC with respect to I_j with a hazard, similarly, has an extra positive term for $\lambda \in [0, 0.5]$. Taken together, it implies a smaller equilibrium q_j and a larger equilibrium I_j that maximize $E[\pi_j]$ than the solutions that maximize π_j^{CE} .

Similarly, we can rewrite equation (18) using $Q^{CE} = (1 - \phi)Q$. The social planner maxi-

mizes the expected welfare.

$$(A10) \quad \max_I \underbrace{\left[(a-b) - \frac{\alpha+\beta}{2} Q^{CE} \right] Q^{CE} - \frac{c+I}{1-\phi} Q^{CE}}_{CE} - \underbrace{\frac{\alpha+\beta}{2N} \frac{\phi}{1-\phi} (Q^{CE})^2}_{Penalty},$$

where the term $-\frac{\alpha+\beta}{2N} \frac{\phi}{1-\phi} (Q^{CE})^2$ reflects a quadratic penalty due to operational uncertainty relative to W^{CE} .

When taking the FOC against I , the penalty produces an extra term whose sign is determined by

$$\frac{\partial \phi(1-\phi)Q(I)^2}{\partial I}$$

This produces

$$\phi'(1-2\phi)Q + 2\phi(1-\phi)Q_I.$$

Again, the term $\phi'(1-2\phi)Q < 0$ for $\lambda \in [0, 0.5]$. It is the direct effect of increasing I as it reduces ϕ and hence volatility in realizing the output. The other term is an indirect effect via the individual firm's decision on the output. This effect has an ambiguous sign as shown in ratio (A14). When $Q_I < 0$, the whole term is negative, and the penalty produces an extra positive term. Thus, the equilibrium I would be higher than that in the certainty-equivalent case.

For an industry that perfectly coordinates in risk management, the penalty term would be $-\frac{\alpha+\beta}{N} \frac{\phi}{1-\phi} (Q^{CE})^2$. The role of the penalty in shaping the equilibrium investment, therefore, is similar to the case of a social planner. The mathematical derivation is hence omitted here.

Analytical Solutions for Simulation

For simulations, we assume linear demand and supply functions. Because elasticity varies with the equilibrium output, we need to rewrite FOCs, Q^* , and I^* as functions without elasticity.

Specifically, we rewrite Equation (13) as

$$P^r + \frac{\partial P^r}{\partial Q} \Phi \frac{q}{Q} Q + \frac{\partial P^r}{\partial Q} (1 - \Phi) \frac{\partial Q/Q}{\partial q/q} Q = P^f + \frac{\partial P^f}{\partial Q} \Phi \frac{q}{Q} Q + \frac{\partial P^f}{\partial Q} (1 - \Phi) \frac{\partial Q/Q}{\partial q/q} Q + \frac{c+I}{1-\Phi},$$

where $\frac{\partial P^r}{\partial Q} = -\alpha$ and $\frac{\partial P^f}{\partial Q} = \beta$. Thus, this condition translates to

$$(A11) \quad a - b - \frac{c+I}{1-\Phi} - \left\{ \alpha \left[(1-\Phi)(1+\xi) + \frac{2\Phi}{N} \right] + \beta \left[(1-\Phi)(1+\theta) + \frac{2\Phi}{N} \right] \right\} Q = 0.$$

Other notation follows the manuscript.

Similarly, we rewrite condition (14) as

$$(A12) \quad (a-b)\Phi_I - (\alpha + \beta) \left[(1-\Phi)(\Phi_I + \Phi_{II}) + \frac{\Phi\Phi_I}{N} \right] Q + 1 = 0,$$

where other notation is consistent with the manuscript.

The equilibrium output for the social planner becomes

$$(A13) \quad Q^* = \frac{a - b - \frac{c+I}{1-\Phi}}{\alpha \left[(1-\Phi)(1+\xi) + \frac{2\Phi}{N} \right] + \beta \left[(1-\Phi)(1+\theta) + \frac{2\Phi}{N} \right]}.$$

Equation (A13) implies that the sign of Q_I is ambiguous. It is because I affects equilibrium output through two opposing channels: a “cost effect” that lowers output and a “reliability effect” that raises output. Investment raises the marginal cost through $\frac{c+I}{1-\Phi}$. A higher I hence reduces Q . Investment also reduces the shutdown probability via $\Phi_I < 0$. This reduces the effective marginal cost by raising $1 - \Phi$ in $\frac{c+I}{1-\Phi}$, inducing a higher Q . A larger I also means a larger denominator because $\frac{2}{N} < 1$, suggesting a smaller Q .

When $|\Phi_I|$ is large, namely, risk is highly responsive to investment, and ξ and θ are large (i.e., close to monopoly-monopsony), the increase in the denominator dominates. Thus, Q_I tends to be negative. Intuitively, when risk-reduction is effective and the firm has a strong ability to control output, the marginal benefit of investing in risk-reduction rises faster than the marginal cost due to

higher risk-reduction efficiency and growth in markups.

When $|\Phi_I|$, ξ , and θ are small, the “cost effect” dominates, and Q_I also tends to be negative. When $|\Phi_I|$ is large, but ξ and θ are small, the “reliability effect” may dominate, leaving $Q_I > 0$ (i.e., the effective marginal cost falls).

The partial derivative, $Q_I = \frac{\partial Q}{\partial I}$, in equation (19) becomes

$$(A14) \quad \frac{AD - BC}{D^2},$$

where $A = -\frac{1}{1-\Phi} - \frac{(c+I)\Phi_I}{(1-\Phi)^2}$, $B = -\left[\alpha(1+\xi) + \beta(1+\theta) - \frac{2(\alpha+\beta)}{N}\right]\Phi_I$, $C = a - b - \frac{c+I}{1-\Phi}$, and $D = \alpha\left[(1-\Phi)(1+\xi) + \frac{2\Phi}{N}\right] + \beta\left[(1-\Phi)(1+\theta) + \frac{2\Phi}{N}\right]$.

An Alternative Modeling Setup

In this appendix section, we discuss alternative modeling settings and the sensitivity of the baseline simulation outcomes.

In particular, if we let the government decide risk-reducing investment as well as firm output, the planner achieves the first-best solution. The planner's problem becomes:

$$(B1) \quad \max_{I, Q} \underbrace{(a-b)(1-\phi(I))Q - (\alpha+\beta) \left[(1-\phi(I))^2 + \frac{(1-\phi(I))\phi(I)}{N} \right] Q^2 - cQ - IQ}_{E[\Pi]} + \underbrace{\frac{\alpha+\beta}{2} \left[(1-\phi(I))^2 + \frac{(1-\phi(I))\phi(I)}{N} \right] Q^2}_{E[CS]+E[PS]}.$$

Given the specification of ϕ for analytical solutions, the two FOCs follow:

$$(B2) \quad \begin{aligned} (a-b)(1-\Phi) - (\alpha+\beta) \left[(1-\Phi)^2 + \frac{(1-\Phi)\Phi}{N} \right] Q &= c + I; \\ -(a-b)\Phi_I + (\alpha+\beta) \left[(1-\Phi)\Phi_I - \frac{(1-2\Phi)\Phi_I}{2N} \right] Q &= 1. \end{aligned}$$

Here $\Phi = \lambda e^{-\gamma(I+\kappa I^2)}$, and $\Phi_I = -\lambda \gamma(1+2\kappa I)e^{-\gamma(I+\kappa I^2)}$.

The equilibrium I and Q are displayed in [Figure B1](#). Although the patterns match panel (b) of [Figure 1](#), the equilibrium Q solved from this alternative social planner problem is weakly higher throughout the range of λ . If there is high market power, the equilibrium Q solved from the alternative setup would be even larger than that from the baseline setup of government decision because the social planner who controls output would further eliminate quantity distortion due to market power. Thus, the incremental CS and PS would be larger in this alternative setup, and so would the incremental social welfare. The difference in ΔW under the two alternative setups is displayed in panel (b) of [Figure B1](#).

In this alternative setup, however, the profits earned by agribusiness firms could be negative, when the CS-plus-PS increment rises faster than the decrease in firm profits. Thus, increasing

social welfare sometimes comes with decreasing and negative firm profits. While this combination of welfare changes is theoretically sound, it is not practical because negative variable profits could imply firm shutdown or even exit. Our two-stage setup in the baseline, which ensures firm profit-maximization, rules out negative profits and hence is more realistic.

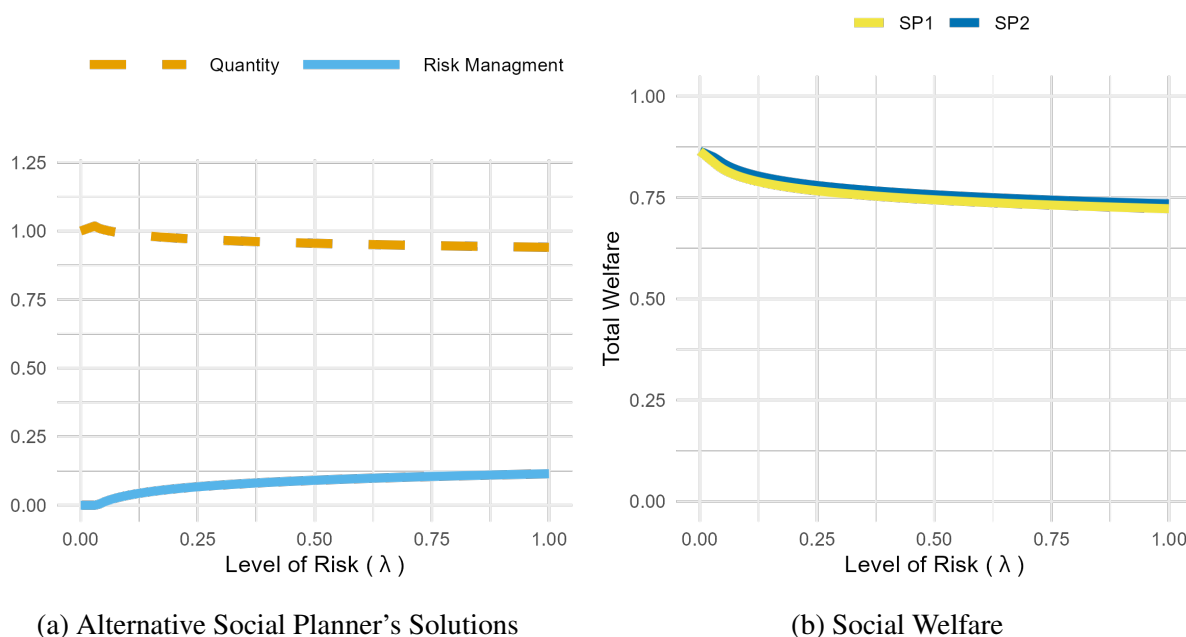


Figure B1: Social planner solutions: Choosing both I and Q

Note: Figure displays results from numerical simulations with respect to the hazard λ . The yellow curve in panel (b) represents the total welfare under the baseline setup, and the blue curve represents the total welfare under the alternative setup of social planner problem described in the appendix. The welfare terms are measured in expectation. All other parameters are set at average values from the distributions described in Table 1.

Parameterizing the Risk Technology

The effectiveness of risk technology in mitigating risk exposure and the size of shocks is an important determinant of the amount that firms invest in equilibrium. In our model, this technology is captured by the term γ . It can be interpreted as a *semi-elasticity of risk management* or the percentage reduction in risk for every unit of risk management invested, mathematically, $-\frac{d\phi}{dI} \frac{1}{\phi}$ is equal to γ if $\kappa = 0$. For example, a value of γ equals 10 implies that moving from \$0.01 to \$0.02 in risk expenditure per dollar of revenue yields a 9.5% reduction in risk.

We pin down the plausible range of γ by referring to statistics of hog diseases and the corresponding vaccines. We choose this context because it is one of the most valuable livestock industries and also because it is where the richest information on diseases and treatments is available. The values of γ lie in the range of 15 to 55 in most cases (i.e., between the upper and lower quartiles). We, therefore, make (15, 55) the baseline range for γ in the simulations.

Table C1: Parameter Values and Disease Information

ϕ_0	ϕ_1	dI	c_{vaccine}	V^r	γ	Disease	Animal	Source
0.91	0.69	0.011	3.90	346.67	15.1	PEDV	piglet	vaccine price: https://www.jrgsupply.com/products/pedv-porcine-epidemic-diarrhea-virus-vaccine-50-dose
0.91	0.69	0.012	4.20	346.67	14.0	PEDV	piglet	vaccine price: https://www.fbn.com/direct/product/porcine-epidemic-diarrhea-vaccine
0.91	0.69	0.013	4.50	346.67	13.0	PEDV	piglet	disease: (Gerdtz and Zakhartchouk 2017)
0.59	0.45	0.011	3.90	346.67	14.8	PEDV	piglet	vaccine price: https://www.jrgsupply.com/products/pedv-porcine-epidemic-diarrhea-virus-vaccine-50-dose

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Table C1 – Continued

ϕ_0	ϕ_1	dI	c_{vaccine}	V^r	γ	Disease	Animal	Source
0.59	0.45	0.012	4.20	346.67	13.7	PEDV	piglet	vaccine price: https://www.fbn.com/direct/product/porcine-epidemic-diarrhea-vaccine
0.59	0.45	0.013	4.50	346.67	12.8	PEDV	piglet	disease: (Gerdtz and Zakhartchouk 2017)
0.057	0.0397	0.005	1.56	346.67	47.2	PRRSV	wean-to-finish	vaccine price: https://www.fbn.com/direct/product/ingelvac-prrs-mlv
0.057	0.0397	0.006	2.06	346.67	35.8	PRRSV	wean-to-finish	disease: (Angulo et al. 2023)
0.057	0.0397	0.007	2.56	346.67	28.8	PRRSV	wean-to-finish	
0.065	0.050	0.005	1.56	346.67	35.9	PRRSV	wean-to-finish	vaccine price: https://www.fbn.com/direct/product/ingelvac-prrs-mlv
0.065	0.050	0.006	2.06	346.67	27.2	PRRSV	wean-to-finish	disease: (Angulo et al. 2023)
0.065	0.050	0.007	2.56	346.67	21.9	PRRSV	wean-to-finish	
0.061	0.030	0.005	2.06	346.67	57.9	PRRSV	finishing	https://www.fbn.com/direct/product/ingelvac-prrs-mlv
0.061	0.030	0.005	2.56	346.67	46.6	PRRSV	finishing	disease: (Renken et al. 2021)
0.061	0.030	0.005	3.06	346.67	39.0	PRRSV	finishing	
0.070	0.029	0.004	1.56	346.67	90.76	PCV2	finishing	vaccine price: https://www.merck-animal-health-usa.com/species/swine/products/circumvent-cml
0.070	0.029	0.006	2.06	346.67	68.72	PCV2	finishing	vaccine price: https://www.scahealth.com/p/ingelvac-circoflex
0.070	0.029	0.007	2.56	346.67	55.29	PCV2	finishing	disease: (Kristensen, Baadsgaard, and Toft 2011)

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Table C1 – Continued

ϕ_0	ϕ_1	dI	c_{vaccine}	V^r	γ	Disease	Animal	Source
0.104	0.050	0.004	1.56	346.67	80.4	PCV2	finishing	vaccine price: https://www.merck-animal-health-usa.com/species/swine/products/circumvent-cml
0.104	0.050	0.006	2.06	346.67	60.9	PCV2	finishing	vaccine price: https://www.scahealth.com/p/ingelvac-circoflex
0.104	0.050	0.007	2.56	346.67	49.0	PCV2	finishing	disease: (Neumann et al. 2009)
0.056	0.030	0.004	1.44	346.67	78.3	PCV2	finishing	vaccine price: https://www.valleyvet.com/ct_detail.html?pgguid=e0b04305-0569-4f04-a233-5e7f1c72705b
0.056	0.030	0.006	1.94	346.67	58.1	PCV2	finishing	
0.056	0.030	0.007	2.44	346.67	46.2	PCV2	finishing	disease: (Jacela et al. 2007)

Note: Authors' creation. Average weight of retail pork per head, q^r , and the retail price of pork, P^r , are obtained from USDA ERS: <https://www.ers.usda.gov/data-products/meat-price-spreads>. The retail value of a hog is $V^r = q^r \times P^r$. The cost of vaccine, c_{vaccine} , is computed by the price of vaccine (e.g., \$100 for 50 doses) and the number of doses needed per head. The normalized cost of vaccine, dI , equals c_{vaccine} divided by v^r . The mortality rate of hogs without vaccine is ϕ_0 and becomes ϕ_1 with vaccine. Because of imperfect implementation, the reduction of mortality rate is always lower in practice than in laboratory. We use the scalar, $\chi_{\text{lab-farm}} = 70\%$, to capture the loss of vaccine effectiveness based on experts' opinions. We compute $\gamma = -\frac{d\phi}{dI} \frac{1}{\phi_0}$ where $d\phi = -(\phi_1 - \phi_0)$.

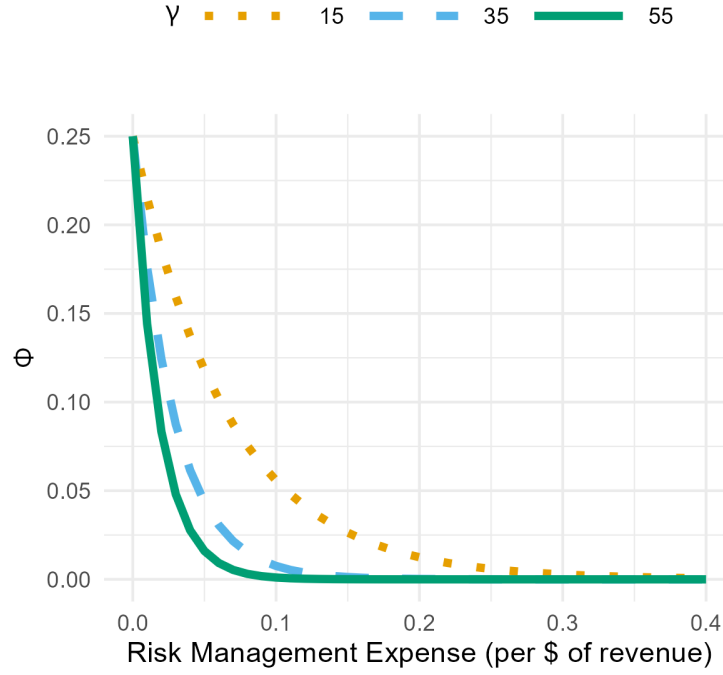


Figure C1: Risk exposure with different risk technologies (γ)

Note: Figure displays $\Phi(I|\gamma, \lambda)$ against I for $\lambda = 0.25$ and different values of γ .

Figure C1 demonstrates the effectiveness of investing in risk management at different values of γ , 15 (lower end of the range), 35 (mean of the range), and 55 (upper end of the range). Higher values of γ lead to steeper declines in $\Phi(I)$ as investment increases, reflecting greater effectiveness of the investment (i.e., higher semi-elasticity of risk management).

Additional Tables and Figures

We include supplemental tables and figures in this appendix section. [Table D1](#) reports starting values, convergence rates, corner solution rates, and the share of draws that result in a positive public-private wedge for the simulation exercise. These values reflect the baseline, perfectly competitive case, and corner solution rates and the sign of ΔI may vary over market-structure values. In column (3), we show that the Newton–Raphson algorithm fails to converge in many cases if the initial values are chosen inappropriately. Our baseline starting values, (0.67, 0.01), are selected to reduce numerical failures rather than arbitrarily.

[Figure D1](#) shows that buyer and seller power both reduce ΔI , with buyer power imposing a steeper decline: θ interacts with the upstream supply elasticity, which exceeds the downstream demand elasticity.

The baseline results hold across alternative farm shares $f \in \{0.20, 0.30, 0.40\}$ and firm counts $N \in \{2, 4, 6\}$ (see [Figure D2](#) (a)–(b)). Smaller f narrows ΔI because firms capture more surplus and internalize a larger share of the investment return, while fewer firms enlarge ΔI because each firm’s market impact is amplified. Exponential and logistic hazard specifications produce nearly identical results over the range of I arising in equilibrium ([Figure D2\(c\)](#)).

[Figure D3](#) decomposes the welfare consequences: the planner’s higher investment generates a large consumer surplus gain, but reduces firm profits, so the net welfare improvement is more modest — the two effects work in opposite directions. Firm profits are omitted from figures in the main manuscript, but can be deduced since $\Delta W = \Delta CS + \Delta \Pi + \Delta PS$.

Table D1: Summary Statistics of Newton-Raphson Solutions at the Competitive Baseline

	Baseline $\lambda \sim N(0.25, 0.075)$	Low Risk $\lambda \sim N(0.10, 0.075)$	Poor Starting Values $\lambda \sim N(0.25, 0.075)$
<i>Solution Method</i>			
Starting value, private (Q, I)	(0.67, 0.01)	(0.67, 0.01)	(0.02, 0.90)
Starting value, public (I)	0.01	0.01	0.90
<i>Total Non-Convergence Rate</i>			
Private	1.1%	20.2%	59.2%
Public	0.6%	14.7%	100.0%
<i>Corner Solution Rate</i>			
Private	1.1%	20.2%	1.1%
Public	0.6%	14.7%	0.6%
<i>Newton-Raphson Failure Rate</i>			
Private	0.0%	0.0%	58.1%
Public	0.0%	0.0%	99.4%
<i>Sign of ΔI</i>			
$\Delta I > 0$ (public > private)	99.4%	85.3%	—
$\Delta I < 0$ (public < private)	0.0%	0.0%	—
$\Delta I = 0$ (both at corner)	0.6%	14.7%	—

Note: Results from 1,000 Monte-Carlo replications per column. All parameters drawn stochastically. $\eta \sim U(0.49, 0.96)$, $\varepsilon \sim U(0.8, 1.2)$, $\gamma \sim N(30, 10)$ truncated at 5. A draw is classified as a corner solution ($I^* = 0$) when the first-order condition for I^* , with the closed-form solution for Q^* (equation 16), is evaluated at $I = 0$ and satisfies the Karush-Kuhn-Tucker condition for a constrained optimum. Otherwise, the interior solution is located via Newton-Raphson, and failure to converge from a given starting value is recorded as a Newton-Raphson Failure. ΔI computed over draws with interior solutions for both problems.

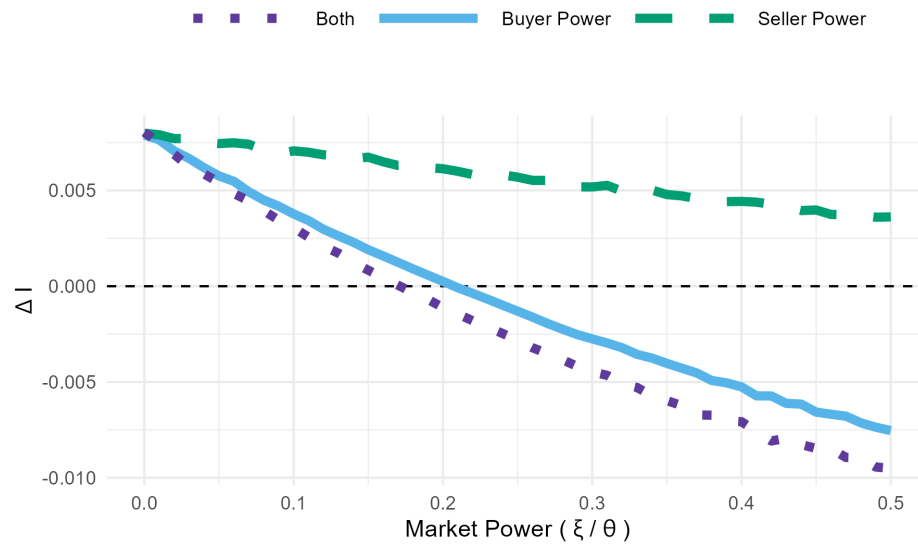


Figure D1: Independently varying buyer and seller power parameters on risk externality

Note: Figures display the wedge in risk investment ΔI plotted over different values of buyer and seller power. The dotted line reflects ΔI when varying them simultaneously over the range of 0 to 0.5. All other parameter values for the simulations are set at baseline levels in [Table 1](#).

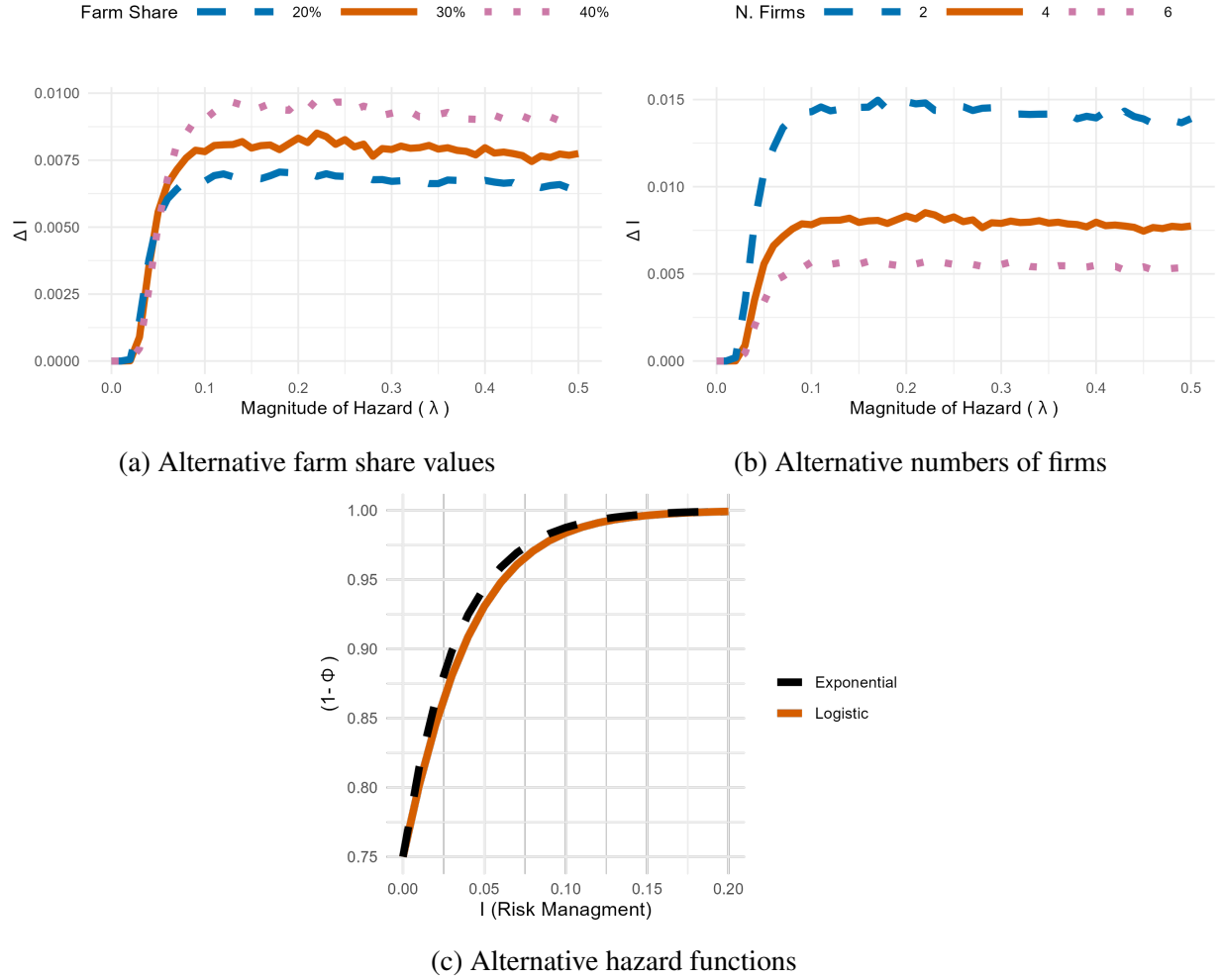


Figure D2: Robustness of risk-investment wedges and hazard functions

Note: Panels (a) and (b) display the wedge in risk investment ΔI plotted over different values of hazard level λ . Panel (a) varies farm share f , while Panel (b) varies the number of firms, N . Panel (c) displays the survival probability, $1 - \Phi$, under different levels of risk management I , comparing an exponential hazard function with a logistic function. All other parameter values for the simulations are set at baseline levels in [Table 1](#).

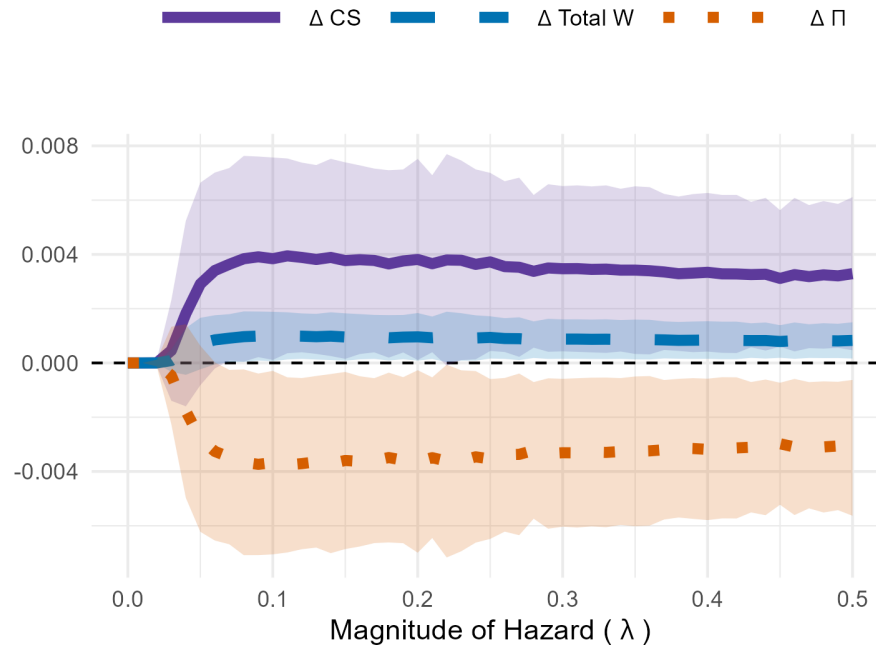


Figure D3: Gaps in CS, profits, and welfare

Note: Figure displays mean and 1.96-standard-deviation bands from 1,000 numerical simulations with respect to the size of hazard that firms face λ . Δ indicates the difference between the social planner's solutions and the private firm's, CS is consumer surplus, W is the social welfare, and Π is aggregate firm profits. All other parameters are set at baseline values as described in [Table 1](#).

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